

Sid Mathur's Thematic ETF Screener – Daily Report

2026-08-25

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Action Center**Run date:** 2026-08-25 **Universe:** 659 ETFs / 26 clusters**Full PC1:** LONG since 2026-08-04 (D+14, -0.0%) **Clusters long:** 12/26 **ETFs long:** 352/659 (53%)**Today's events (cluster level)**

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

SELL (1)

cluster	theme	spread_sell	p_h5	nss
cluster_12	China Equity	-0.35	0.88	0.046

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_10	Healthcare	7.738	1.000	1.00	1.000	1.00
cluster_8	U.S. Dividend Equity	4.347	1.000	1.00	1.000	0.99
cluster_4	U.S. Large Cap Value	2.855	1.000	1.00	0.877	0.73
cluster_3	U.S. Mid Cap Equity	0.822	0.983	0.74	0.071	0.01
cluster_13	Financials	2.881	1.000	1.00	0.068	0.15

Imminent SELL (top 2 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_6	U.S. Large Cap Growth	-0.328	0.522	0.41	-0.394	0.0
cluster_19	Consumer Staples	-0.029	0.562	0.57	-0.211	-0.3

Full Sample PC1



Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: LONG since 2026-08-04 (14 days, -0.0%)

5dma: 145.250 20dma: 144.619 50dma: 143.098

Spread vs 50dma: +2.151 Spread vs 20dma: +0.631

P(buy crossover within 5 days): 99.9% P(sell): 13.7%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
SRLN	cluster_23	Bank Loan / Floating Rate	1.000	5	0.1%	6.01%	0.89
FLOT	cluster_23	Bank Loan / Floating Rate	1.000	5	0.1%	5.88%	0.89
GUNR	cluster_11	Materials	1.000	13	4.9%	5.61%	0.98
BKLN	cluster_23	Bank Loan / Floating Rate	1.000	5	0.1%	4.30%	0.89
FLRN	cluster_23	Bank Loan / Floating Rate	1.000	5	0.1%	2.45%	0.89
IAU	cluster_17	Gold	1.000	11	9.4%	2.31%	0.95
SLV	cluster_17	Gold	1.000	11	9.4%	1.26%	0.95
AMLPL	cluster_11	Materials	1.000	13	4.9%	1.00%	0.98
URA	cluster_11	Materials	1.000	13	4.9%	0.70%	0.98
FLTR	cluster_23	Bank Loan / Floating Rate	1.000	5	0.1%	0.68%	0.89
GLD	cluster_17	Gold	1.000	11	9.4%	0.08%	0.95
PDBC	cluster_22	Broad Commodities	0.961	10	3.7%	5.23%	0.98
IGV	cluster_15	Technology	0.941	14	0.6%	4.69%	0.90
ICSH	cluster_24	Cash / Ultra-Short Fixed In...	0.931	5	0.0%	7.96%	0.86
SHV	cluster_24	Cash / Ultra-Short Fixed In...	0.931	5	0.0%	5.09%	0.86
MINT	cluster_24	Cash / Ultra-Short Fixed In...	0.931	5	0.0%	3.47%	0.86
GBIL	cluster_24	Cash / Ultra-Short Fixed In...	0.931	5	0.0%	2.51%	0.86
IEMG	cluster_7	Emerging Markets Equity	0.920	8	0.0%	4.73%	0.99
VEA	cluster_2	Developed ex-US Equity	0.906	17	1.8%	8.90%	0.99
VOO	cluster_1	U.S. Large Cap Equity	0.552	13	-1.2%	8.21%	0.99
EWJ	cluster_21	Japan Equity	0.275	12	-0.0%	5.34%	0.97
QQQ	cluster_6	U.S. Large Cap Growth	0.003	10	-3.1%	4.15%	0.98
XLP	cluster_19	Consumer Staples	-0.303	4	0.7%	5.46%	0.99
XLU	cluster_19	Consumer Staples	-0.303	4	0.7%	3.37%	0.99

Watchlist (imminent-buy clusters, p > 80%, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_13	Financials	1.00	0.150	1.094	0.96
VTV	cluster_4	U.S. Large Cap Value	1.00	0.733	1.092	0.97

VHT	cluster_10	Healthcare	1.00	1.000	1.088	0.93
SCHD	cluster_8	U.S. Dividend Equity	1.00	0.986	0.891	0.96
BWX	cluster_25	International Fixed Income	1.00	0.994	0.752	0.93
INDA	cluster_18	India Equity	1.00	0.639	0.641	0.91
NOBL	cluster_18	India Equity	1.00	0.639	0.078	0.91
PPLT	cluster_18	India Equity	1.00	0.639	0.029	0.91
KWEB	cluster_12	China Equity	0.92	0.046	0.561	0.92

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
10	cluster_10	Healthcare	Healthcare	17	FLAT	shifted			1.000	+1.335
11	cluster_11	Materials	Materials	17	LONG	shifted	13	4.9%	1.000	-0.000
17	cluster_17	Gold	Gold	12	LONG		11	9.4%	1.000	+0.000
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	LONG		5	0.1%	1.000	+0.000
25	cluster_25	International Fixed Income	International Fix...	6	FLAT				0.994	+0.004
8	cluster_8	U.S. Dividend Equity	U.S. Dividend Equ...	25	FLAT	shifted			0.986	+0.033
22	cluster_22	Broad Commodities	Broad Commodities	9	LONG	shifted	10	3.7%	0.961	+0.198
15	cluster_15	Technology	Technology	14	LONG	shifted	14	0.6%	0.941	+1.886
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	LONG		5	0.0%	0.931	+0.006
7	cluster_7	Emerging Markets Equity	Emerging Markets ...	30	LONG	shifted	8	0.0%	0.920	+0.126
2	cluster_2	Developed ex-US Equity	Developed ex-US E...	79	LONG	shifted	17	1.8%	0.906	+0.344
4	cluster_4	U.S. Large Cap Value	U.S. Large Cap Va...	60	FLAT	shifted			0.733	-0.231
18	cluster_18	India Equity	India Equity	12	FLAT	shifted			0.639	+1.221
1	cluster_1	U.S. Large Cap Equity	U.S. Large Cap Eq...	81	LONG		13	-1.2%	0.552	-0.299
21	cluster_21	Japan Equity	Japan Equity	9	LONG	shifted	12	-0.0%	0.275	+0.457
13	cluster_13	Financials	Financials	14	FLAT	shifted			0.150	-0.181
12	cluster_12	China Equity	China Equity	15	FLAT	shifted			0.046	-0.954
3	cluster_3	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	72	FLAT	shifted			0.006	-0.872
6	cluster_6	U.S. Large Cap Growth	U.S. Large Cap Gr...	38	LONG	shifted	10	-3.1%	0.003	-0.777
20	cluster_20	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT	shifted			-0.039	-0.612
5	cluster_5	Aggregate Fixed Income	Aggregate Fixed I...	58	FLAT				-0.098	+0.207
9	cluster_9	High Yield Bond	High Yield Bond	23	FLAT	shifted			-0.169	-1.125
19	cluster_19	Consumer Staples	Consumer Staples	11	LONG	shifted	4	0.7%	-0.303	-1.260
16	cluster_16	Municipal Bond	Municipal Bond	13	FLAT				-0.534	-0.020
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT	shifted			-0.701	+0.183
14	cluster_14	Real Estate	Real Estate	14	FLAT	shifted			-0.831	-1.074

Cluster Details

Cluster 1 – U.S. Large Cap Equity

Members: 81 **Var explained:** 0.858 **Signal:** LONG since 2026-08-05 (13 days, -1.2%)

Signal logic:

- Latest buy_cross: 2026-08-05 – filter **PASS** (252d at date = +20.1%)
- Latest sell_cross: 2026-07-17
- Current 252d return: +18.2% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-07-17)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1191
U.S. Total Market Equity	7	0.0930
Global Equity	7	0.0910
Technology	5	0.0601
Consumer Discretionary	5	0.0587
Asset Allocation	4	0.0483
Industrials	4	0.0408
U.S. Equity	3	0.0388

U.S. Mega Cap Equity	3	0.0387
U.S. Multifactor Equity	3	0.0387
U.S. Quality / Factor Equity	3	0.0382
U.S. Mid Cap Growth	3	0.0381
U.S. Momentum Equity	3	0.0344
Communication Services	3	0.0336
U.S. ESG Equity	2	0.0264
Momentum Equity	2	0.0238
Convertible Bond	2	0.0227
Commodity	1	0.0132
Large Cap Value Equities	1	0.0131
Large Cap Blend Equities	1	0.0131
Dividend Equity	1	0.0128
U.S. Equal Weight Equity	1	0.0126
U.S. Dividend Equity	1	0.0125
Growth Equity	1	0.0123
IPO Equity	1	0.0118
U.S. Mid Cap Momentum Equity	1	0.0117
Alternatives	1	0.0115
U.S. Small Cap Momentum Equity	1	0.0112
International IPO Equity	1	0.0101
Israel Equity	1	0.0097

Replication basket – VOO (cor=0.99) basket lead at lag 1: -0.009

ticker	rep_wt	name	category	etf_signal
VOO	1.007	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
50	IWV	0.0133	iShares Russell 3000 Fund	U.S. Total Market Equity	LONG
51	IYY	0.0133	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	LONG
57	IWB	0.0133	iShares Russell 1000 ETF	U.S. Large Cap Equity	LONG
62	VTHR	0.0133	Vanguard Russell 3000 ETF	U.S. Total Market Equity	LONG
65	VONE	0.0133	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	LONG
69	ITOT	0.0133	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	LONG
71	SCHB	0.0133	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	LONG
72	SCHX	0.0133	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	LONG
75	VTI	0.0133	Vanguard Total Stock Market ETF	U.S. Total Market Equity	LONG
5	SPY	0.0132	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	LONG

12	URTH	0.0132	iShares MSCI World ETF	Global Equity	LONG
37	TILT	0.0132	FlexShares Morningstar US Market Factors...	U.S. Multifactor Equity	FLAT
46	JHML	0.0132	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	LONG
49	SUSA	0.0132	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	LONG
54	SPYX	0.0132	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	LONG
56	ESGU	0.0132	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	LONG
63	GSLC	0.0132	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	LONG
67	VV	0.0132	Vanguard Large-Cap ETF	U.S. Large Cap Equity	LONG
68	ILCB	0.0132	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	LONG
70	IVV	0.0132	iShares Core S&P 500 ETF	U.S. Large Cap Equity	LONG
73	SPTM	0.0132	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	LONG
74	VOO	0.0132	Vanguard S&P 500 ETF	U.S. Large Cap Equity	LONG
7	ACWI	0.0131	iShares MSCI ACWI ETF	Global Equity	LONG
13	CRBN	0.0131	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	LONG
14	SPGM	0.0131	State Street SPDR Portfolio MSCI Global...	Global Equity	LONG
15	VT	0.0131	Vanguard Total World Stock Index ETF	Global Equity	LONG
48	DSI	0.0131	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	LONG
52	LRGF	0.0131	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	LONG
58	IWL	0.0131	iShares Russell Top 200 ETF	U.S. Equity	LONG
64	EPS	0.0131	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	LONG
66	MGC	0.0131	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	LONG
8	AOA	0.0130	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	LONG
53	OEF	0.0129	iShares S&P 100 Fund	U.S. Mega Cap Equity	LONG
59	QUAL	0.0129	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	FLAT
36	IMCG	0.0128	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
44	QDF	0.0128	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
45	CATH	0.0128	Global X S&P 500 Catholic Values ETF	Global Equity	LONG
35	VOT	0.0127	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
43	IOO	0.0127	iShares Global 100 ETF	Global Equity	LONG
55	XLG	0.0127	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	LONG
60	QUS	0.0127	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT
11	AOR	0.0126	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	LONG
18	QQEW	0.0126	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
19	XT	0.0126	iShares Future Exponential Technologies...	Technology	LONG
28	IWP	0.0126	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
61	SPHQ	0.0126	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
21	FTC	0.0125	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
42	SPHB	0.0125	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
47	DGRW	0.0125	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	LONG
81	OMFL	0.0124	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
26	RPG	0.0123	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
4	ROBO	0.0122	ROBO Global Robotics and Automation Ind...	Technology	FLAT
24	IYC	0.0122	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	LONG
22	PDP	0.0121	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
31	VCR	0.0120	Vanguard Consumer Discretion ETF	Consumer Discretionary	LONG

34	FDIS	0.0120	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	LONG
77	VFMO	0.0120	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
3	BOTZ	0.0119	Global X Robotics & Artificial Intellig...	Technology	FLAT
1	XLY	0.0118	State Street Consumer Discretionary Sel...	Consumer Discretionary	LONG
10	AOM	0.0118	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	LONG
23	FPX	0.0118	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
16	FV	0.0117	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
27	XMMO	0.0117	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
25	CWB	0.0116	State Street SPDR Bloomberg Convertible...	Convertible Bond	LONG
6	QAI	0.0115	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	LONG
30	MTUM	0.0115	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
32	VOX	0.0113	Vanguard Communication Services ETF	Communication Services	LONG
33	FCOM	0.0113	Fidelity MSCI Communication Services In...	Communication Services	LONG
38	DWAS	0.0112	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
29	ICVT	0.0111	iShares Convertible Bond ETF	Convertible Bond	LONG
17	PRNT	0.0110	3D Printing (The) ETF	Technology	LONG
2	XLC	0.0109	State Street Communication Services Sel...	Communication Services	LONG
9	AOK	0.0109	iShares Core 30/70 Conservative Allocat...	Asset Allocation	LONG
80	PRN	0.0109	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
76	QMOM	0.0108	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	LONG
79	RTH	0.0107	VanEck Retail ETF	Consumer Discretionary	FLAT
39	PPA	0.0102	Invesco Aerospace & Defense ETF	Industrials	FLAT
41	XAR	0.0102	State Street SPDR S&P Aerospace & Defen...	Industrials	FLAT
20	FPXI	0.0101	First Trust International Equity Opport...	International IPO Equity	FLAT
78	EIS	0.0097	iShares MSCI Israel ETF	Israel Equity	LONG
40	ITA	0.0095	iShares U.S. Aerospace & Defense ETF	Industrials	FLAT

Cluster 2 – Developed ex-US Equity

Members: 79 **Var explained:** 0.841 **Signal:** LONG since 2026-07-30 (17 days, 1.8%)

Signal logic:

- Latest buy_cross: 2026-07-30 – filter **PASS** (252d at date = +20.0%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +19.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-07-30 > last sell at 2026-06-24)



Category breakdown

category	n_members	abs_weight
International Equity	21	0.2771
Developed ex-US Equity	9	0.1219
International Dividend Equity	6	0.0770
Global ex-US Equity	5	0.0676
Europe Equity	4	0.0536
Alternatives	4	0.0483
Eurozone Equity	3	0.0386
Asia-Pacific Equity	2	0.0250
Real Estate	2	0.0232
Global Equity	2	0.0227
Currency	2	0.0185
Commodity	1	0.0137
Global ex-US Small Cap Equity	1	0.0133
Germany Equity	1	0.0128
France Equity	1	0.0128
United Kingdom Equity	1	0.0126
Italy Equity	1	0.0126
Europe Financials	1	0.0124
Sweden Equity	1	0.0124
Australia Equity	1	0.0123
Netherlands Equity	1	0.0123
Spain Equity	1	0.0120

Canada Equity	1	0.0118
Switzerland Equity	1	0.0117
Austria Equity	1	0.0116
Belgium Equity	1	0.0115
Singapore Equity	1	0.0108
Poland Equity	1	0.0108
Consumer Discretionary	1	0.0097
New Zealand Equity	1	0.0094

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
VEA	0.911	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
65	IEFA	0.0138	iShares Core MSCI EAFE ETF	Developed ex-US Equity	LONG
3	EFA	0.0137	iShares MSCI EAFE ETF	Developed ex-US Equity	LONG
4	VEA	0.0137	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	LONG
51	GSIE	0.0137	Goldman Sachs ActiveBeta International ...	Commodity	LONG
54	ESGD	0.0137	iShares ESG Aware MSCI EAFE ETF	International Equity	LONG
66	SCHF	0.0137	Schwab International Equity ETF	Developed ex-US Equity	LONG
67	SPDW	0.0137	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	LONG
23	DWM	0.0136	WisdomTree International Equity Fund	International Dividend E...	LONG
36	TLTD	0.0136	FlexShares Morningstar Developed Market...	International Equity	LONG
43	INTF	0.0136	iShares International Equity Factor ETF	International Equity	LONG
61	IXUS	0.0136	iShares Core MSCI Total International S...	Global ex-US Equity	LONG
7	IEV	0.0135	iShares Europe ETF	Europe Equity	LONG
30	PXF	0.0135	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	LONG
37	JPIN	0.0135	JPMorgan Diversified Return Internation...	International Equity	LONG
40	ACWX	0.0135	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	LONG
41	CWI	0.0135	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	LONG
44	IQLT	0.0135	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
46	QEFA	0.0135	State Street SPDR MSCI EAFE StrategicFa...	International Equity	LONG
50	FNDF	0.0135	Schwab Fundamental International Equity...	International Equity	LONG
60	IEUR	0.0135	iShares Core MSCI Europe ETF	Europe Equity	LONG
62	VEU	0.0135	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	LONG
63	VGK	0.0135	Vanguard FTSEEuropean ETF	Europe Equity	LONG
64	VXUS	0.0135	Vanguard Total International Stock ETF	Global ex-US Equity	LONG
33	SCZ	0.0134	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	LONG

34	EFV	0.0134	iShares MSCI EAFE Value ETF	Developed ex-US Equity	LONG
35	FNDC	0.0134	Schwab Fundamental International Small ...	International Equity	LONG
48	RODM	0.0134	Hartford Multifactor Developed Markets ...	International Equity	LONG
49	VYMI	0.0134	Vanguard International High Dividend Yi...	International Dividend E...	LONG
58	SCHC	0.0134	Schwab International Small-Cap Equity E...	Developed ex-US Equity	LONG
21	EZU	0.0133	iShares MSCI Eurozone ETF	Eurozone Equity	LONG
26	IQDF	0.0133	FlexShares International Quality Divide...	International Equity	LONG
59	VSS	0.0133	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	LONG
9	DLS	0.0132	WisdomTree International SmallCap Fund	International Equity	LONG
31	EFG	0.0132	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	LONG
45	IVLU	0.0132	iShares MSCI Intl Value Factor ETF	International Equity	LONG
71	HFXI	0.0132	NYLI FTSE International Equity Currency...	International Equity	LONG
5	FEP	0.0131	First Trust Europe AlphaDEX Fund	Europe Equity	LONG
47	FEZ	0.0131	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	LONG
56	VIGI	0.0131	Vanguard International Dividend Appreci...	International Dividend E...	LONG
42	IMTM	0.0130	iShares MSCI Intl Momentum Factor ETF	International Equity	LONG
32	GWX	0.0129	State Street SPDR S&P International Sma...	International Equity	LONG
2	EWG	0.0128	iShares MSCI Germany Index Fund	Germany Equity	LONG
6	VIDI	0.0128	Vident International Equity Strategy ETF	International Equity	LONG
17	EWQ	0.0128	iShares MSCI France Index Fund	France Equity	LONG
76	IPKW	0.0128	Invesco International BuyBack Achievers...	International Equity	LONG
22	IDV	0.0127	iShares International Select Dividend E...	International Dividend E...	LONG
19	EWU	0.0126	iShares MSCI United Kingdom ETF	United Kingdom Equity	LONG
55	HDEF	0.0126	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	LONG
79	EWI	0.0126	iShares MSCI Italy ETF	Italy Equity	FLAT
24	EPP	0.0125	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	LONG
52	IDLV	0.0125	Invesco S&P International Developed Low...	International Equity	LONG
78	BBAX	0.0125	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	LONG
25	EUFN	0.0124	iShares MSCI Europe Financials ETF	Europe Financials	LONG
70	EWD	0.0124	iShares MSCI Sweden ETF	Sweden Equity	LONG
14	EWA	0.0123	iShares MSCI Australia Index Fund	Australia Equity	LONG
38	DBEF	0.0123	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
39	HEFA	0.0123	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
53	EFAV	0.0123	iShares MSCI EAFE Min Vol Factor ETF	International Equity	LONG
74	EWN	0.0123	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
13	HEZU	0.0122	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	LONG
29	DWX	0.0122	State Street SPDR S&P International Div...	International Dividend E...	LONG
16	EWP	0.0120	iShares MSCI Spain ETF	Spain Equity	FLAT
11	IHDG	0.0119	WisdomTree International Hedged Quality...	Alternatives	FLAT
12	PID	0.0119	Invesco International Dividend Achiever...	International Dividend E...	FLAT
10	HEDJ	0.0118	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
20	EWG	0.0118	iShares MSCI Canada Index Fund	Canada Equity	LONG
57	VNQI	0.0118	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	LONG
15	EWL	0.0117	iShares MSCI Switzerland ETF	Switzerland Equity	LONG
27	NFRA	0.0117	FlexShares STOXX Global Broad Infrastru...	Global Equity	LONG

73	EW0	0.0116	iShares MSCI Austria ETF	Austria Equity	LONG
77	EWK	0.0115	iShares MSCI Belgium ETF	Belgium Equity	LONG
8	RWX	0.0114	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
28	IGF	0.0111	iShares Global Infrastructure ETF	Global Equity	FLAT
18	EWS	0.0108	iShares MSCI Singapore ETF	Singapore Equity	LONG
69	EPOL	0.0108	iShares MSCI Poland ETF	Poland Equity	LONG
68	FXA	0.0100	Invesco CurrencyShares Australian Dolla...	Currency	LONG
1	ESPO	0.0097	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
72	ENZL	0.0094	iShares MSCI New Zealand ETF	New Zealand Equity	LONG
75	FXC	0.0085	Invesco CurrencyShares Canadian Dollar ...	Currency	LONG

Cluster 3 – U.S. Mid Cap Equity

Members: 72 **Var explained:** 0.897 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +40.0%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +23.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
U.S. Small Cap Equity	10	0.1440
U.S. Mid Cap Equity	6	0.0863
U.S. Small Cap Value	6	0.0859
U.S. Small Cap Growth	6	0.0841
Consumer Discretionary	4	0.0517
Industrials	4	0.0498
U.S. Mid Cap Value	3	0.0432
U.S. Mid Cap Growth	3	0.0420
Leveraged / Inverse	2	0.0276
Dividend Equity	2	0.0273
U.S. Small Cap Low Volatility	2	0.0261
U.S. Small Cap Fundamental Equity	1	0.0146
U.S. Small Cap ESG Equity	1	0.0145
Unknown	1	0.0145
U.S. Mid/Small Cap Equity	1	0.0145
U.S. Small/Mid Cap Fundamental Equity	1	0.0145
Mid Cap Value Equities	1	0.0144
U.S. Small Cap Multifactor Equity	1	0.0144
Small Cap Value Equities	1	0.0142
U.S. Small/Mid Cap Equity	1	0.0142
U.S. Mid Cap Dividend Equity	1	0.0141
U.S. ESG Equity	1	0.0141
U.S. Extended Market Equity	1	0.0141
U.S. Small Cap Dividend Equity	1	0.0139
Value Equity	1	0.0139
U.S. Equity	1	0.0139
U.S. Mid Cap Quality Equity	1	0.0139
Growth Equity	1	0.0139
U.S. Small Cap Industrials	1	0.0136
U.S. Small Cap Momentum Equity	1	0.0136
U.S. Micro Cap Equity	1	0.0134
U.S. Small Cap Consumer Discretionary	1	0.0131
U.S. Dividend Equity	1	0.0131
U.S. Mid Cap Low Volatility	1	0.0124
Real Estate	1	0.0113

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.013

ticker	rep_wt	name	category	etf_signal
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IJH 1.015 iShares Core S&P Mid-Cap ETF U.S. Mid Cap Equity FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
47	FNDA	0.0146	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
8	FNX	0.0145	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
11	NUSC	0.0145	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
12	PRFZ	0.0145	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
15	MDY	0.0145	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
23	IVOO	0.0145	Vanguard S&P Mid-Cap 400 ETF		FLAT
24	VIOO	0.0145	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT
27	IJR	0.0145	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
30	IJH	0.0145	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
31	SPMD	0.0145	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
32	SPSM	0.0145	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
33	VB	0.0145	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
59	VBR	0.0145	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
4	IWM	0.0144	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	LONG
9	FYX	0.0144	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
13	SMLF	0.0144	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	LONG
16	IJT	0.0144	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
18	VIOG	0.0144	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
21	SLYG	0.0144	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
25	VTWO	0.0144	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	LONG
34	SCHA	0.0144	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
44	EZM	0.0144	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
50	IWN	0.0144	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
52	IJJ	0.0144	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
54	IVOV	0.0144	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
55	MDYV	0.0144	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
5	JHMM	0.0143	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
7	SCHM	0.0143	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
53	IJS	0.0143	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
56	SLYV	0.0143	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
58	VTWV	0.0143	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
65	RWK	0.0143	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
43	EES	0.0142	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
45	PSC	0.0142	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
57	VIOV	0.0142	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
61	RWJ	0.0142	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
71	SDVY	0.0142	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	FLAT
72	ISMD	0.0142	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	FLAT

29	VXF	0.0141	Vanguard Extended Market ETF	U.S. Extended Market Equ...	LONG
42	DON	0.0141	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
60	ETHO	0.0141	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
17	IJK	0.0140	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
19	IVOG	0.0140	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
20	MDYG	0.0140	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
28	ISCG	0.0139	iShares Morningstar Small-Cap Growth ETF	Growth Equity	LONG
41	DES	0.0139	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
63	XMHQ	0.0139	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
64	RFV	0.0139	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
66	OMFS	0.0139	Invesco Russell 2000 Dynamic Multifact...	U.S. Equity	FLAT
70	FYT	0.0139	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
14	IWO	0.0137	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
22	VTWG	0.0137	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
67	RZV	0.0137	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
26	VBK	0.0136	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	LONG
62	PSCI	0.0136	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
69	XSMO	0.0136	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
35	FXD	0.0135	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	FLAT
10	IWC	0.0134	iShares Microcap ETF	U.S. Micro Cap Equity	LONG
39	RSPD	0.0132	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	FLAT
51	SMMV	0.0132	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
6	REGL	0.0131	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	FLAT
37	FTXR	0.0131	First Trust Nasdaq Transportation ETF	Industrials	FLAT
40	SMDV	0.0131	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
68	PSCD	0.0131	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	FLAT
46	XTN	0.0129	State Street SPDR S&P Transportation ETF	Industrials	FLAT
49	XSLV	0.0128	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
1	XRT	0.0126	State Street SPDR S&P Retail ETF	Consumer Discretionary	FLAT
38	IYT	0.0124	iShares U.S. Transportation ETF	Industrials	LONG
48	XMLV	0.0124	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	FLAT
36	PEJ	0.0123	Invesco Leisure and Entertainment ETF	Consumer Discretionary	FLAT
2	JETS	0.0114	U.S. Global Jets ETF	Industrials	FLAT
3	REM	0.0113	iShares Mortgage Real Estate ETF	Real Estate	LONG

Cluster 4 – U.S. Large Cap Value

Members: 60 **Var explained:** 0.890 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +26.6%)

- Latest sell_cross: 2026-07-23
- Current 252d return: +20.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Value	10	0.1728
U.S. Dividend Equity	8	0.1360
Industrials	7	0.1142
Materials	4	0.0627
U.S. Mid Cap Equity	3	0.0516
U.S. Multifactor Equity	3	0.0512
U.S. Value Factor Equity	3	0.0502
Utilities	3	0.0468
U.S. Fundamental Equity	2	0.0348
U.S. Equal Weight Equity	2	0.0347
U.S. Mid Cap Value	2	0.0346
U.S. Equity	2	0.0343
Consumer Discretionary	2	0.0272
Large Cap Blend Equities	1	0.0173
U.S. Size Factor Equity	1	0.0172
U.S. Revenue Weighted Equity	1	0.0171
Value Equity	1	0.0170
U.S. Mega Cap Value	1	0.0169

U.S. Buyback Equity	1	0.0168
U.S. Large Cap Equity	1	0.0167
U.S. Moat Equity	1	0.0162
Communication Services	1	0.0137

Replication basket – VTV (cor=0.97) basket lead at lag 1: -0.003

ticker	rep_wt	name	category	etf_signal
VTV	1.092	Vanguard Value ETF	U.S. Large Cap Value	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
3	RSP	0.0175	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
26	IWD	0.0175	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
36	VONV	0.0175	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
15	PRF	0.0174	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
18	CFA	0.0174	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
22	FNDX	0.0174	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
29	JPUS	0.0174	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
42	SCHV	0.0174	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
48	IWS	0.0174	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
6	FEX	0.0173	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
39	ILCV	0.0173	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
41	IUSV	0.0173	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
59	JPME	0.0173	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT
23	EQAL	0.0172	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
27	IWR	0.0172	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
28	IVE	0.0172	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
30	SIZE	0.0172	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	FLAT
34	VOOV	0.0172	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	FLAT
40	IMCB	0.0172	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
43	SPYV	0.0172	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
45	VTV	0.0172	Vanguard Value ETF	U.S. Large Cap Value	LONG
49	VOE	0.0172	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
52	DTD	0.0172	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
16	RWL	0.0171	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
20	NULV	0.0171	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
44	VO	0.0171	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
54	DGRO	0.0171	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
60	JVAL	0.0171	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG

12	IYJ	0.0170	iShares U.S. Industrials ETF	Industrials	FLAT
17	AIVL	0.0170	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
24	IWX	0.0170	iShares Russell Top 200 Value ETF	Value Equity	LONG
25	ONEV	0.0170	State Street SPDR Russell 1000 Low Vola...	U.S. Equity	FLAT
38	VYM	0.0170	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
50	PFM	0.0170	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
51	DLN	0.0170	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
55	VIG	0.0170	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
37	MGV	0.0169	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	FLAT
5	PKW	0.0168	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
10	RDVY	0.0168	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
21	FDVV	0.0168	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
4	FXR	0.0167	First Trust Industrials AlphaDEX	Industrials	FLAT
53	DIA	0.0167	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	FLAT
33	VIS	0.0166	Vanguard Industrials ETF	Industrials	FLAT
35	FIDU	0.0166	Fidelity MSCI Industrials Index ETF	Industrials	FLAT
1	XLI	0.0164	State Street Industrial Select Sector S...	Industrials	FLAT
19	CFO	0.0164	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
11	MOAT	0.0162	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
56	PAVE	0.0162	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
31	VLUE	0.0161	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
9	FIW	0.0160	First Trust Water ETF	Utilities	FLAT
32	VAW	0.0160	Vanguard Materials ETF	Materials	LONG
2	XLB	0.0158	State Street Materials Select Sector SP...	Materials	LONG
7	PHO	0.0158	Invesco Water Resources ETF	Utilities	FLAT
14	RSPM	0.0156	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
13	IYM	0.0152	iShares U.S. Basic Materials ETF	Materials	LONG
8	CGW	0.0150	Invesco S&P Global Water Index ETF	Utilities	FLAT
58	PKB	0.0148	Invesco Building & Construction ETF	Industrials	FLAT
47	XHB	0.0142	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	FLAT
57	IYZ	0.0137	iShares U.S. Telecommunications ETF	Communication Services	LONG
46	ITB	0.0130	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG

Cluster 5 – Aggregate Fixed Income

Members: 58 **Var explained:** 0.831 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +1.3% – filter **ALLOWS** new entries today

- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
Fixed Income	19	0.3284
Investment Grade Corporate Bond	13	0.2303
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0891
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0305
Agency Fixed Income	1	0.0166
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.005

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	FLAT
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	FLAT
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	FLAT
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	FLAT
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	FLAT
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	FLAT
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
11	GVI	0.0182	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	FLAT
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	FLAT
27	USIG	0.0181	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	FLAT
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	FLAT
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	FLAT
52	SPMB	0.0180	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	FLAT
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	FLAT
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	FLAT
10	CORP	0.0178	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	FLAT
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	FLAT
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	FLAT
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	FLAT
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
16	SPLB	0.0172	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	FLAT
44	SPTL	0.0172	State Street SPDR Portfolio Long Term T...	Fixed Income	FLAT
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	FLAT
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	FLAT

39	AGZ	0.0166	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	FLAT
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	FLAT
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	LONG
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	LONG
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	LONG
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	LONG
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	FLAT
33	BSCQ	0.0155	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	LONG
14	IBDR	0.0150	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	LONG
38	CMBS	0.0135	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	LONG
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 6 – U.S. Large Cap Growth

Members: 38 **Var explained:** 0.893 **Signal:** LONG since 2026-08-10 (10 days, -3.1%)

Signal logic:

- Latest buy_cross: 2026-08-10 – filter **PASS** (252d at date = +36.3%)
- Latest sell_cross: 2026-07-16
- Current 252d return: +33.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-10 > last sell at 2026-07-16)



Category breakdown

category	n_members	abs_weight
Technology	19	0.4990
U.S. Large Cap Growth	11	0.2989
Growth Equity	2	0.0536
U.S. Equity	1	0.0273
U.S. Mega Cap Growth	1	0.0270
Options Income	1	0.0246
U.S. Small Cap Technology	1	0.0243
Trend-Following Equity	1	0.0242
U.S. Trend-Following Equity	1	0.0211

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.029

ticker	rep_wt	name	category	etf_signal
QQQ	1.033	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

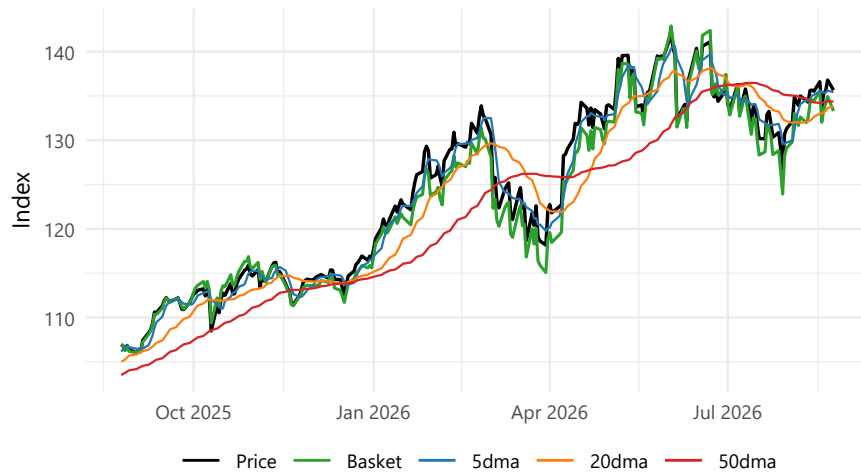
	ticker	PC_wt	name	category	etf_signal
16	IGM	0.0276	iShares Expanded Tech Sector ETF	Technology	LONG
5	QQQ	0.0275	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	LONG
18	IYW	0.0275	iShares U.S. Technology ETF	Technology	LONG
24	VGT	0.0275	Vanguard Information Tech ETF	Technology	LONG
26	FTEC	0.0275	Fidelity MSCI Information Technology In...	Technology	LONG
1	XLK	0.0274	State Street Technology Select Sector S...	Technology	LONG
29	ILCG	0.0274	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
17	IXN	0.0273	iShares Global Tech ETF	Technology	LONG
20	ONEQ	0.0273	Fidelity Nasdaq Composite Index ETF	U.S. Equity	LONG
22	IWF	0.0272	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	LONG
23	IVW	0.0272	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
25	VOOG	0.0272	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	LONG
27	VONG	0.0272	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	LONG
30	IUSG	0.0272	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	LONG
32	SPYG	0.0272	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	LONG
12	XNTK	0.0271	State Street SPDR NYSE Technology ETF	Technology	LONG
33	VUG	0.0271	Vanguard Growth ETF	U.S. Large Cap Growth	LONG
19	NULG	0.0270	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
21	IWY	0.0270	iShares Russell Top 200 Growth ETF	Growth Equity	LONG
28	MGK	0.0270	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	LONG
7	QTEC	0.0269	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
31	SCHG	0.0268	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	LONG
4	AIQ	0.0266	Global X Artificial Intelligence & Tech...	Technology	LONG
11	FXL	0.0266	First Trust Technology AlphaDEX	Technology	LONG
15	PWB	0.0266	Invesco Large Cap Growth ETF	Growth Equity	LONG
3	SMH	0.0259	VanEck Semiconductor ETF	Technology	FLAT
9	TDIV	0.0259	First Trust NASDAQ Technology Dividend ...	Technology	LONG
2	SOXX	0.0256	iShares PHLX SOX Semiconductor Sector I...	Technology	FLAT
38	QTUM	0.0255	Defiance Quantum ETF	Technology	LONG
10	XSD	0.0250	State Street SPDR S&P Semiconductor ETF	Technology	FLAT
36	FTXL	0.0250	First Trust Nasdaq Semiconductor ETF	Technology	FLAT
8	PSI	0.0249	Invesco Semiconductors ETF	Technology	FLAT
6	NXTG	0.0248	First Trust Indxx NextG ETF	Technology	LONG
14	QYLD	0.0246	Global X NASDAQ 100 Covered Call ETF	Options Income	LONG
35	PSCT	0.0243	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	FLAT
37	PTF	0.0243	Invesco Dorsey Wright Technology Moment...	Technology	LONG
13	PTNQ	0.0242	Pacer Trendpilot 100 ETF	Trend-Following Equity	LONG
34	PTLC	0.0211	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	LONG

Cluster 7 – Emerging Markets Equity

Members: 30 **Var explained:** 0.798 **Signal:** LONG since 2026-08-12 (8 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-08-12 – filter **PASS** (252d at date = +29.5%)
- Latest sell_cross: 2026-07-07
- Current 252d return: +28.7% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-12 > last sell at 2026-07-07)

**Category breakdown**

category	n_members	abs_weight
Emerging Markets Equity	19	0.6753
Asia Equity	2	0.0703
Asia ex-Japan Equity	1	0.0361
Taiwan Equity	1	0.0313
Global Dividend Equity	1	0.0299
South Africa Equity	1	0.0297
South Korea Equity	1	0.0294
Latin America Equity	1	0.0275
Mexico Equity	1	0.0245
Brazil Equity	1	0.0233

Chile Equity	1	0.0227
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Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.031

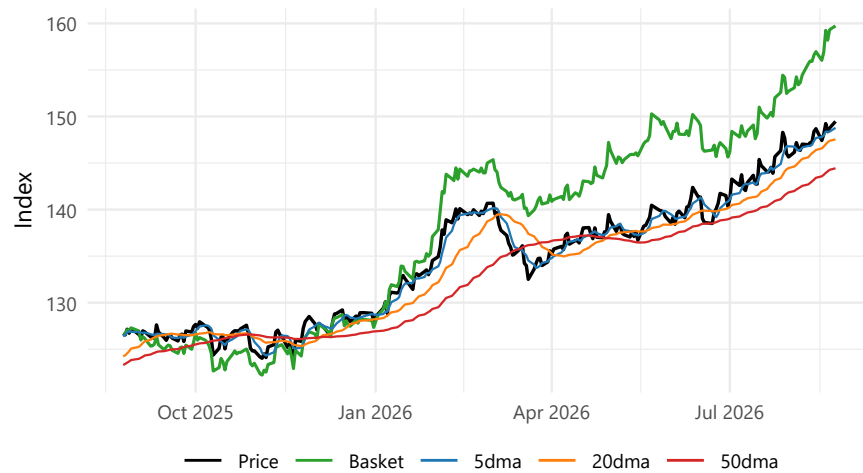
ticker	rep_wt	name	category	etf_signal
IEMG	0.908	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
22	IEMG	0.0370	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	LONG
3	EEM	0.0369	iShares MSCI Emerging Index Fund	Emerging Markets Equity	LONG
24	SPEM	0.0369	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	LONG
4	VWO	0.0368	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
17	GEM	0.0368	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	LONG
21	ESGE	0.0368	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	LONG
23	SCHE	0.0367	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
19	XSOE	0.0365	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	LONG
16	EMGF	0.0364	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	LONG
6	AAXJ	0.0361	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	LONG
13	PXH	0.0359	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	LONG
18	FNDE	0.0359	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	LONG
12	EEMA	0.0358	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	LONG
15	GMF	0.0356	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	LONG
9	DGS	0.0354	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG
20	EEMV	0.0352	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	LONG
8	DEM	0.0350	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
11	AIA	0.0348	iShares Asia 50 ETF	Asia Equity	LONG
5	FEM	0.0342	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	LONG
7	EWX	0.0342	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
14	DVYE	0.0323	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	LONG
10	EWI	0.0313	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
30	FEMS	0.0308	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	LONG
26	SDIV	0.0299	Global X SuperDividend ETF	Global Dividend Equity	LONG
29	EZA	0.0297	iShares MSCI South Africa Index Fund	South Africa Equity	LONG
2	EWY	0.0294	iShares MSCI South Korea ETF	South Korea Equity	FLAT
28	ILF	0.0275	iShares Latin America 40 ETF	Latin America Equity	FLAT
25	EWI	0.0245	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	EWZ	0.0233	iShares MSCI Brazil ETF	Brazil Equity	FLAT
27	ECH	0.0227	iShares MSCI Chile ETF	Chile Equity	LONG

Cluster 8 – U.S. Dividend Equity**Members:** 25 **Var explained:** 0.868 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-20 – filter **PASS** (252d at date = +20.8%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +18.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-20)

**Category breakdown**

category	n_members	abs_weight
U.S. Dividend Equity	13	0.5283
U.S. Low Volatility Equity	4	0.1562
Value Equity	3	0.1200
Dividend Equity	2	0.0792
U.S. Defensive Equity	1	0.0404
U.S. Quality / Factor Equity	1	0.0390
Global Equity	1	0.0368

Replication basket – SCHD (cor=0.96) basket lead at lag 1: 0.004

ticker	rep_wt	name	category	etf_signal
SCHD	0.891	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SDY	0.0418	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
15	DVY	0.0417	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
2	FVD	0.0415	First Trust VL Dividend	U.S. Dividend Equity	LONG
10	SCHD	0.0413	Schwab US Dividend Equity ETF	U.S. Dividend Equity	LONG
14	SDOG	0.0413	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
11	FTA	0.0412	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
20	SPYD	0.0412	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
5	DHS	0.0411	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
19	ONEY	0.0411	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT
18	SPHD	0.0409	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
6	CDC	0.0404	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
4	FDL	0.0402	First Trust Morningstar ETF	U.S. Dividend Equity	LONG
25	LGLV	0.0401	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
8	LVHD	0.0399	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	FLAT
3	PWV	0.0398	Invesco Large Cap Value ETF	Value Equity	LONG
9	HDV	0.0394	iShares Core High Dividend ETF	U.S. Dividend Equity	LONG
16	RDIV	0.0394	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
12	PEY	0.0393	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
17	RPV	0.0391	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
13	DIV	0.0390	Global X Super Dividend ETF	U.S. Dividend Equity	FLAT
21	FTCS	0.0390	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
22	OUSA	0.0383	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	LONG
23	SPLV	0.0381	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	FLAT
24	USMV	0.0381	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
1	ACWV	0.0368	iShares MSCI Global Min Vol Factor ETF	Global Equity	LONG

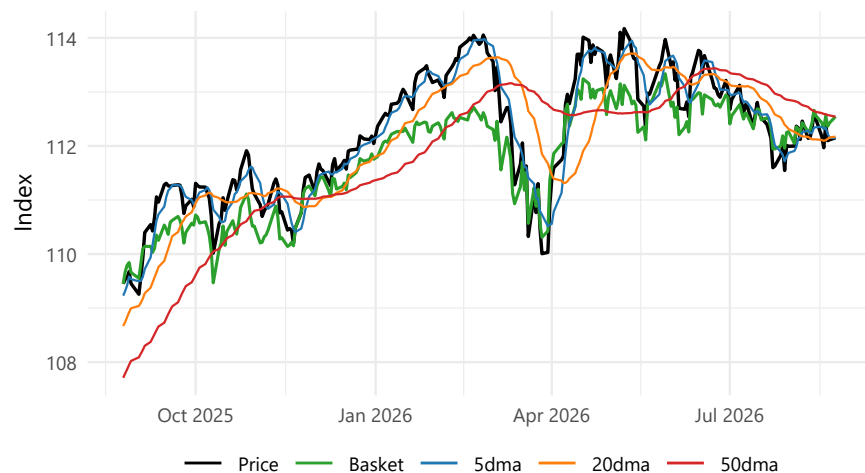
Cluster 9 – High Yield Bond

Members: 23 **Var explained:** 0.738 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.2%)

- Latest sell_cross: 2026-07-07
- Current 252d return: +3.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-07 more recent than last effective buy 2026-06-17)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.4783
Preferred Stock	7	0.2821
Emerging Markets Fixed Income	6	0.2396

Replication basket – HYG (cor=0.94) basket lead at lag 1: 0.045

ticker	rep_wt	name	category	etf_signal
HYG	0.906	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	HYG	0.0488	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG

13	JNK	0.0488	State Street SPDR Bloomberg High Yield ...	High Yield Bond	FLAT
19	HVLB	0.0486	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
18	BBHY	0.0485	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	FLAT
16	SHYG	0.0483	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG
14	SJNK	0.0482	State Street SPDR Bloomberg Short Term ...	High Yield Bond	FLAT
17	FALN	0.0480	iShares Fallen Angels USD Bond ETF	High Yield Bond	FLAT
15	ANGL	0.0479	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	FLAT
6	HYS	0.0474	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	FLAT
9	PFF	0.0443	iShares Preferred and Income Securities...	Preferred Stock	LONG
2	EMB	0.0442	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
23	VWOB	0.0440	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	FLAT
3	HYLS	0.0438	First Trust Tactical High Yield ETF	High Yield Bond	LONG
20	PCY	0.0432	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	FLAT
11	PFXF	0.0424	VanEck Preferred Securities ex Financia...	Preferred Stock	LONG
7	PGX	0.0412	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0407	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0400	First Trust Preferred Securities and In...	Preferred Stock	FLAT
5	PGF	0.0397	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0369	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	FLAT
21	EBND	0.0365	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
22	EMLC	0.0348	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
8	VRP	0.0339	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 10 – Healthcare

Members: 17 **Var explained:** 0.756 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +34.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9419
U.S. Small Cap Healthcare	1	0.0581

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.006

ticker	rep_wt	name	category	etf_signal
VHT	1.088	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0656	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0655	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0648	First Trust Health Care AlphaDEX	Healthcare	FLAT
14	RSPH	0.0641	Invesco S&P 500 Equal Weight Health Car...	Healthcare	FLAT
13	IYH	0.0640	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0625	iShares Global Healthcare ETF	Healthcare	FLAT

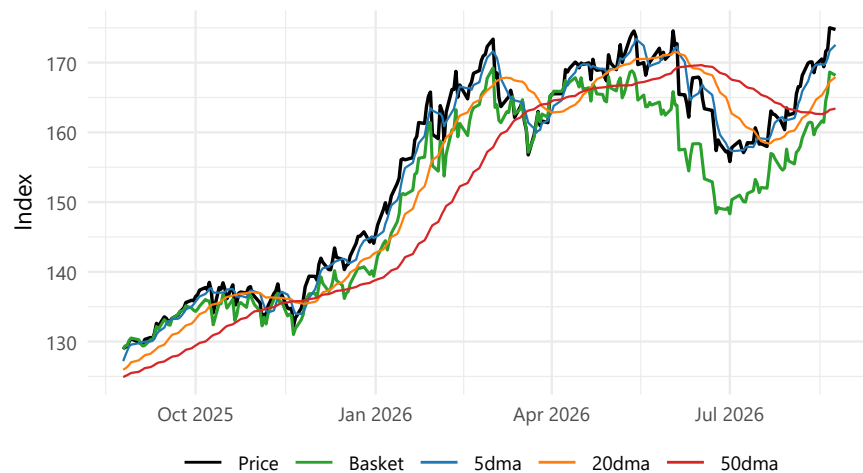
1	XLV	0.0623	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0605	iShares Biotechnology ETF	Healthcare	FLAT
6	FBT	0.0597	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0581	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	FLAT
7	XHE	0.0571	State Street SPDR S&P Health Care Equip...	Healthcare	FLAT
12	IHI	0.0565	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0546	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0537	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0533	State Street SPDR S&P Biotech ETF	Healthcare	FLAT
2	ARKG	0.0507	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0470	iShares U.S. Health Care Providers ETF	Healthcare	FLAT

Cluster 11 – Materials

Members: 17 **Var explained:** 0.682 **Signal:** LONG since 2026-08-05 (13 days, 4.9%)

Signal logic:

- Latest buy_cross: 2026-08-05 – filter **PASS** (252d at date = +34.1%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +38.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-05 > last sell at 2026-06-09)



Category breakdown

category	n_members	abs_weight
Materials	9	0.5600
MLP / Energy Infrastructure	4	0.2164
Energy	1	0.0646
Consumer Staples	1	0.0598
Norway Equity	1	0.0557
Industrial Metals	1	0.0436

Replication basket – AMLP+URA+GUNR (cor=0.98) basket lead at lag 1: 0.047

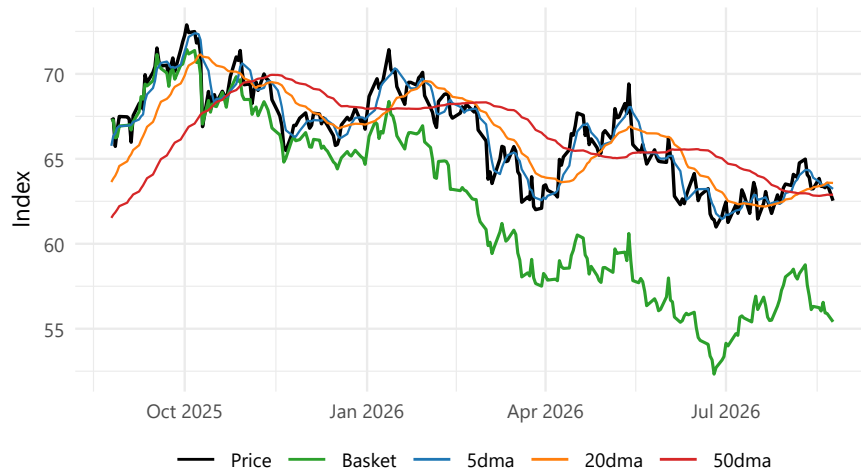
ticker	rep_wt	name	category	etf_signal
AMLP	0.138	Alerian MLP ETF	MLP / Energy Infrastr...	LONG
URA	0.096	Global X Uranium ETF	Materials	LONG
GUNR	0.772	FlexShares Global Upstream Natural Re...	Materials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	GNR	0.0694	State Street SPDR S&P Global Natural Re...	Materials	LONG
10	GUNR	0.0690	FlexShares Global Upstream Natural Reso...	Materials	LONG
17	NANR	0.0670	State Street SPDR S&P North American Na...	Materials	LONG
16	IGE	0.0646	iShares North American Natural Resource...	Energy	LONG
9	MXI	0.0634	iShares Global Materials ETF	Materials	LONG
12	PICK	0.0632	iShares MSCI Global Select Metals & Min...	Materials	LONG
3	XME	0.0613	State Street SPDR S&P Metals & Mining E...	Materials	LONG
13	SLX	0.0608	VanEck Steel ETF	Materials	FLAT
5	MOO	0.0598	VanEck Agribusiness ETF	Consumer Staples	FLAT
2	COPX	0.0591	Global X Copper Miners ETF	Materials	LONG
8	MLPX	0.0560	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
14	NORW	0.0557	Global X MSCI Norway ETF	Norway Equity	LONG
4	EMLP	0.0546	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	LONG
6	AMLP	0.0536	Alerian MLP ETF	MLP / Energy Infrastruct...	LONG
7	MLPA	0.0522	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
1	URA	0.0467	Global X Uranium ETF	Materials	LONG
15	DBB	0.0436	Invesco DB Base Metals Fund	Industrial Metals	LONG

Cluster 12 – China Equity**Members:** 15 **Var explained:** 0.794 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-08-03 – filter **PASS** (252d at date = +2.6%)
- Latest sell_cross: 2026-08-24
- Current 252d return: -3.9% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-08-24 more recent than last effective buy 2026-08-03)

**Category breakdown**

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0644
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.561	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0740	WisdomTree China ex-State-Owned Enterpr...	China Equity	LONG
9	GXC	0.0738	State Street SPDR S&P China ETF	China Equity	LONG
1	MCHI	0.0732	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0712	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0668	Xtrackers Harvest CSI 300 China A-Share...	China Equity	FLAT
8	CNYA	0.0660	iShares MSCI China A ETF	China Equity	FLAT
13	KGRN	0.0653	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0644	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0597	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
12	EWI	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
14	CNXT	0.0587	VanEck ChiNext Innovators ETF	China Equity	FLAT
15	KURE	0.0581	KraneShares MSCI All China Health Care ...	China Equity	LONG

Cluster 13 – Financials

Members: 14 **Var explained:** 0.871 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +13.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.9383
Global Equity	1	0.0617

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.012

ticker	rep_wt	name	category	etf_signal
XLF	1.094	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0757	Vanguard Financials ETF	Financials	FLAT
11	FNCL	0.0756	Fidelity MSCI Financials Index ETF	Financials	LONG
7	IYF	0.0754	iShares U.S. Financial ETF	Financials	FLAT
5	FXO	0.0752	First Trust Financials AlphaDEX	Financials	FLAT
8	IYG	0.0749	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0742	State Street Financial Select Sector SP...	Financials	LONG

9	KBWB	0.0733	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0713	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0707	State Street SPDR S&P Capital Markets E...	Financials	LONG
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	FLAT
13	IAI	0.0695	iShares U.S. Broker-Dealers & Securitie...	Financials	FLAT
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0630	State Street SPDR S&P Insurance ETF	Financials	FLAT
12	PSP	0.0617	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 14 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-08-06
- Current 252d return: +14.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-06 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: -0.001

ticker	rep_wt	name	category	etf_signal
VNQ	0.968	Vanguard Real Estate ETF	Real Estate	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	LONG
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	LONG
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0721	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT
1	XLRE	0.0720	State Street Real Estate Select Sector ...	Real Estate	LONG
5	ICF	0.0720	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0716	iShares Global REIT ETF	Real Estate	LONG
3	RWO	0.0712	State Street SPDR Dow Jones Global Real...	Real Estate	FLAT
4	REZ	0.0696	iShares Residential and Multisector Rea...	Real Estate	LONG
14	NURE	0.0687	Nuveen Short-Term REIT ETF	Real Estate	FLAT
12	INDS	0.0682	Pacer Industrial Real Estate ETF	Real Estate	FLAT

Cluster 15 – Technology

Members: 14 **Var explained:** 0.827 **Signal:** LONG since 2026-08-04 (14 days, 0.6%)

Signal logic:

- Latest buy_cross: 2026-08-04 – filter **PASS** (252d at date = +6.7%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +5.8% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-04 > last sell at 2026-07-23)



Category breakdown

category	n_members	abs_weight
Technology	10	0.7125
Financials	2	0.1429
IPO Equity	1	0.0731
Consumer Discretionary	1	0.0715

Replication basket – IGV (cor=0.90) basket lead at lag 1: 0.054

ticker	rep_wt	name	category	etf_signal
IGV	0.951	iShares Expanded Tech-Software Sector...	Technology	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
12	FDN	0.0747	First Trust DJ Internet Index Fund	Technology	LONG
6	ARKW	0.0745	ARK Next Generation Internet ETF	Technology	LONG
2	SKYY	0.0741	First Trust Cloud Computing ETF	Technology	LONG
14	FINX	0.0740	Global X FinTech ETF	Financials	FLAT

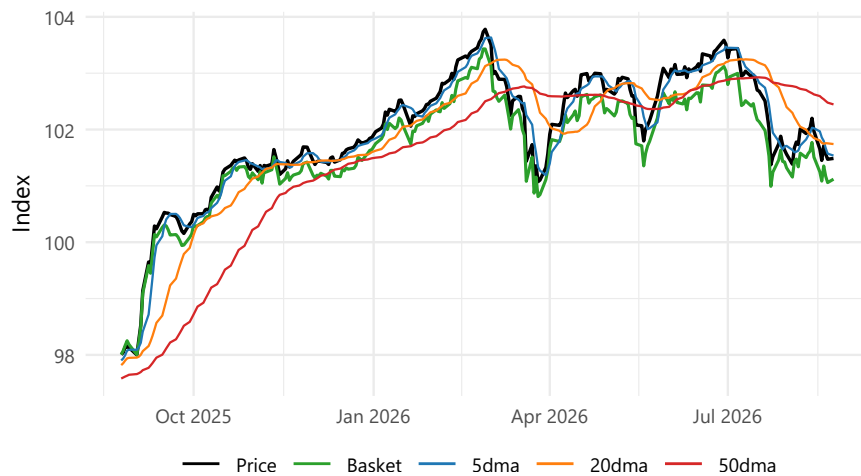
11	PNQI	0.0734	Invesco Nasdaq Internet ETF	Technology	FLAT
10	IPO	0.0731	Renaissance IPO ETF	IPO Equity	LONG
5	ARKK	0.0730	ARK Innovation ETF	Technology	LONG
1	IGV	0.0720	iShares Expanded Tech-Software Sector E...	Technology	FLAT
8	IBUY	0.0715	Amplify Online Retail ETF	Consumer Discretionary	FLAT
9	CIBR	0.0693	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
7	ARKQ	0.0691	ARK Autonomous Technology & Robotics ETF	Technology	LONG
13	IPAY	0.0689	Amplify Digital Payments ETF	Financials	FLAT
4	HACK	0.0683	Amplify Cybersecurity ETF	Technology	FLAT
3	BLOK	0.0641	Amplify Blockchain Technology ETF	Technology	FLAT

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.820 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-08-21
- Current 252d return: +3.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-08-21 more recent than last effective buy 2026-05-28)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.178

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	FLAT
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	FLAT
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	FLAT
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	FLAT
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	FLAT
7	NYF	0.0794	iShares New York Muni Bond ETF	Municipal Bond	FLAT
1	FMB	0.0793	First Trust Managed Municipal ETF	Municipal Bond	FLAT
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	FLAT
4	MUNI	0.0770	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	FLAT
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	FLAT
3	HYMB	0.0722	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	FLAT
12	SUB	0.0693	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	FLAT

Cluster 17 – Gold

Members: 12 **Var explained:** 0.864 **Signal:** LONG since 2026-08-07 (11 days, 9.4%)

Signal logic:

- Latest buy_cross: 2026-08-07 – filter **PASS** (252d at date = +36.2%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +47.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-07 > last sell at 2026-05-20)



Category breakdown

category	n_members	abs_weight
Materials	5	0.4158
Gold	3	0.2503
Precious Metals	2	0.1723
Silver	2	0.1616

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
GLD	0.027	SPDR Gold Shares	Gold	LONG
IAU	0.751	iShares Gold Trust Shares	Gold	LONG
SLV	0.411	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0865	Invesco DB Precious Metals Fund	Precious Metals	LONG
7	GLTR	0.0858	abrdn Physical Precious Metals Basket S...	Precious Metals	LONG

2	GDXJ	0.0841	VanEck Junior Gold Miners ETF	Materials	LONG
1	GDX	0.0840	VanEck Gold Miners ETF	Materials	LONG
3	GLD	0.0835	SPDR Gold Shares	Gold	LONG
9	IAU	0.0835	iShares Gold Trust Shares	Gold	LONG
10	SGOL	0.0833	abrtn Physical Gold Shares ETF	Gold	LONG
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	LONG
6	SIL	0.0829	Global X Silver Miners ETF	Materials	LONG
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	LONG
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0807	abrtn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – India Equity

Members: 12 **Var explained:** 0.339 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-08-03 – filter **BLOCKED** (252d at date = -9.8%; needs > 0)
- Latest effective buy: 2026-05-06 – delayed re-arm from blocked cross (252d at fire = +4.6%)
- Latest sell_cross: 2026-07-28
- Current 252d return: -12.8% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-28 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4522
Malaysia Equity	1	0.1095
Indonesia Equity	1	0.1010
Philippines Equity	1	0.0982
Vietnam Equity	1	0.0731
Turkey Equity	1	0.0590
U.S. Dividend Equity	1	0.0531
Agricultural Commodities	1	0.0255
Platinum	1	0.0232
Natural Gas	1	0.0053

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
NOBL	0.078	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.641	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1520	iShares MSCI India ETF	India Equity	FLAT
3	INDY	0.1504	iShares India 50 ETF	India Equity	FLAT
4	EPI	0.1498	WisdomTree India Earnings Fund	India Equity	FLAT
10	EWM	0.1095	iShares MSCI Malaysia Index Fund	Malaysia Equity	LONG
8	EIDO	0.1010	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
9	EPHE	0.0982	iShares MSCI Philippines ETF	Philippines Equity	FLAT
12	VNM	0.0731	VanEck Vietnam ETF	Vietnam Equity	FLAT
11	TUR	0.0590	iShares MSCI Turkey ETF	Turkey Equity	LONG
5	NOBL	0.0531	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
2	DBA	0.0255	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
6	PPLT	0.0232	abrdn Physical Platinum Shares ETF	Platinum	FLAT
7	UNG	0.0053	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 19 – Consumer Staples**Members:** 11 **Var explained:** 0.740 **Signal:** LONG since 2026-08-18 (4 days, 0.7%)**Signal logic:**

- Latest buy_cross: 2026-08-18 – filter **PASS** (252d at date = +5.0%)
- Latest sell_cross: 2026-08-05
- Current 252d return: +5.1% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-18 > last sell at 2026-08-05)

**Category breakdown**

category	n_members	abs_weight
Consumer Staples	6	0.5743
Utilities	4	0.3503
U.S. Small Cap Consumer Staples	1	0.0753

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.017

ticker	rep_wt	name	category	etf_signal
XLU	0.363	State Street Utilities Select Sector ...	Utilities	FLAT

XLP 0.587 State Street Consumer Staples Select ... Consumer Staples LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0984	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0984	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0976	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0951	iShares U.S. Consumer Staples ETF	Consumer Staples	FLAT
3	KXI	0.0948	iShares Global Consumer Staples ETF	Consumer Staples	FLAT
11	PBJ	0.0900	Invesco Food & Beverage ETF	Consumer Staples	LONG
7	IDU	0.0887	iShares U.S. Utilities ETF	Utilities	FLAT
9	FUTY	0.0874	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
8	VPU	0.0873	Vanguard Utilities ETF	Utilities	FLAT
2	XLU	0.0870	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0753	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

Cluster 20 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.904 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-07-02
- Current 252d return: -0.5% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-02 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.014

ticker	rep_wt	name	category	etf_signal
VTIP	0.267	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.761	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1036	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	FLAT
6	TIPX	0.1035	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1024	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDTT	0.1008	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.1000	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT

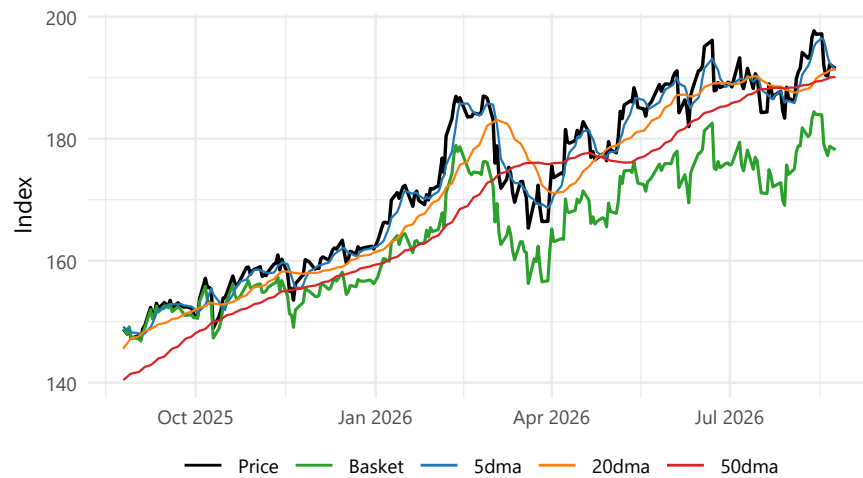
9	STIP	0.0997	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0989	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0884	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 21 – Japan Equity

Members: 9 **Var explained:** 0.887 **Signal:** LONG since 2026-08-06 (12 days, -0.0%)

Signal logic:

- Latest buy_cross: 2026-08-06 – filter **PASS** (252d at date = +35.6%)
- Latest sell_cross: 2026-07-28
- Current 252d return: +29.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-06 > last sell at 2026-07-28)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7805

Asia-Pacific Equity	2	0.2195
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Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.043

ticker	rep_wt	name	category	etf_signal
EWJ	0.907	iShares MSCI Japan Index Fund	Japan Equity	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

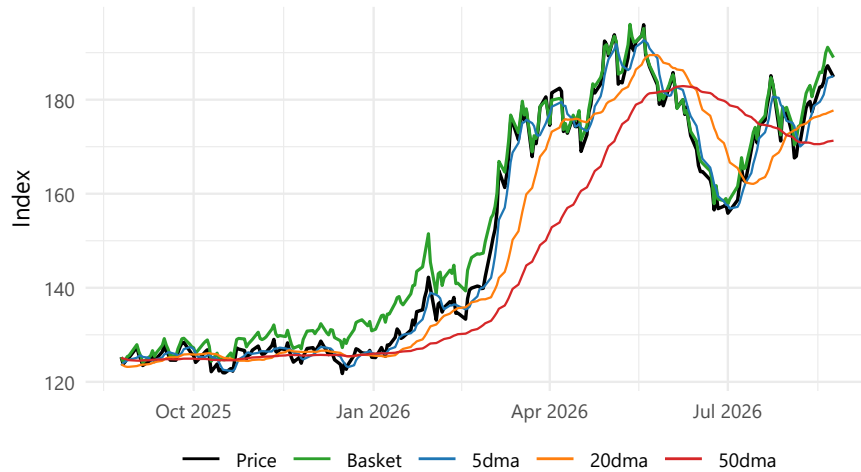
	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	LONG
8	BBJP	0.1142	JPMorgan BetaBuilders Japan ETF	Japan Equity	LONG
6	DBJP	0.1118	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
9	HEWJ	0.1118	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
3	IPAC	0.1114	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1110	Franklin FTSE Japan Hedged ETF	Japan Equity	LONG
2	DXJ	0.1095	WisdomTree Japan Hedged Equity Fund	Japan Equity	LONG
4	VPL	0.1081	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	LONG
7	EWJV	0.1079	iShares MSCI Japan Value ETF	Japan Equity	LONG

Cluster 22 – Broad Commodities

Members: 9 **Var explained:** 0.911 **Signal:** LONG since 2026-08-10 (10 days, 3.7%)

Signal logic:

- Latest buy_cross: 2026-08-10 – filter **PASS** (252d at date = +45.4%)
- Latest sell_cross: 2026-08-06
- Current 252d return: +49.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-10 > last sell at 2026-08-06)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6698
Oil	2	0.2183
Energy Commodities	1	0.1119

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.075

ticker	rep_wt	name	category	etf_signal
PDBC	1.181	Invesco Optimum Yield Diversified Com...	Broad Commodities	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	GSG	0.1152	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
2	DBC	0.1151	Invesco DB Commodity Index Tracking Fund	Broad Commodities	FLAT
7	PDBC	0.1150	Invesco Optimum Yield Diversified Commo...	Broad Commodities	FLAT
8	COMT	0.1145	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
9	DBE	0.1119	Invesco DB Energy Fund	Energy Commodities	FLAT

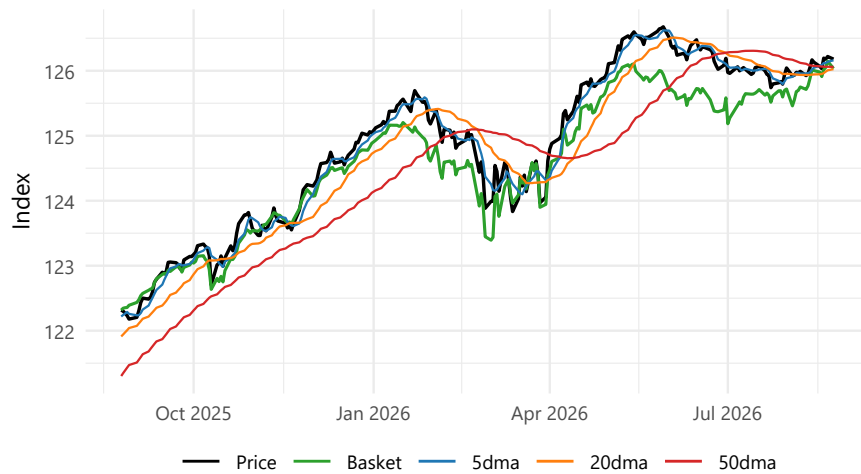
1	USO	0.1098	United States Oil Fund	Oil	LONG
4	DBO	0.1084	Invesco DB Oil Fund	Oil	LONG
6	DJP	0.1055	iPath Bloomberg Commodity Index Total R...	Broad Commodities	FLAT
3	FTGC	0.1045	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.555 **Signal:** LONG since 2026-08-17 (5 days, 0.1%)

Signal logic:

- Latest buy_cross: 2026-08-17 – filter **PASS** (252d at date = +3.2%)
- Latest sell_cross: 2026-07-23
- Current 252d return: +3.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-17 > last sell at 2026-07-23)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3618
Interest Rate Hedged Fixed Income	3	0.3245

Floating Rate Fixed Income 3 0.3137

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.294	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	LONG
BKLN	0.215	Invesco Senior Loan ETF	Bank Loan / Floating ...	LONG
SRLN	0.300	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	LONG
FLRN	0.123	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	LONG
FLTR	0.034	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

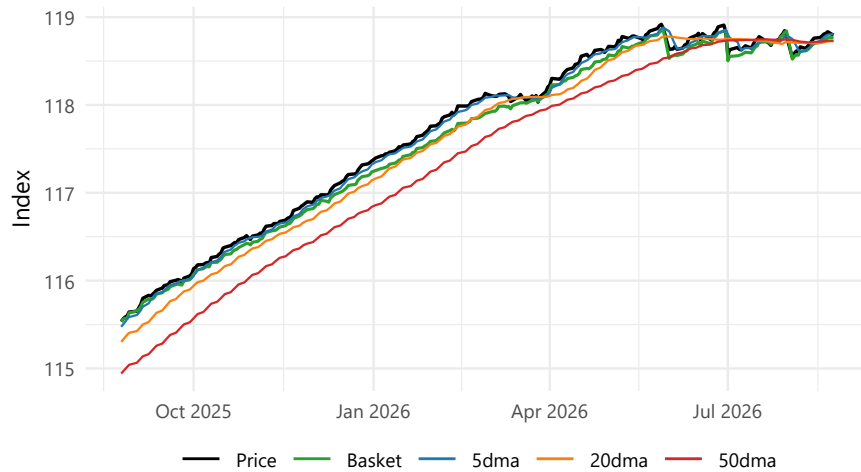
	ticker	PC_wt	name	category	etf_signal
2	SRLN	0.1251	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	LONG
3	BKLN	0.1207	Invesco Senior Loan ETF	Bank Loan / Floating Rate	LONG
1	FTSL	0.1160	First Trust Senior Loan Fund	Bank Loan / Floating Rate	LONG
5	LQDH	0.1143	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1085	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	LONG
6	IGBH	0.1055	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1047	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1040	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	LONG
9	FLTR	0.1012	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	LONG

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 **Var explained:** 0.459 **Signal:** LONG since 2026-08-17 (5 days, 0.0%)

Signal logic:

- Latest buy_cross: 2026-08-17 – filter **PASS** (252d at date = +3.0%)
- Latest sell_cross: 2026-08-05
- Current 252d return: +2.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-08-17 > last sell at 2026-08-05)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.86) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
SHV	0.254	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	LONG
MINT	0.173	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	LONG
GBIL	0.126	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	LONG
ICSH	0.398	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1330	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	LONG
6	ICSH	0.1311	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	LONG
2	MINT	0.1160	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	LONG
4	NEAR	0.1101	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	LONG

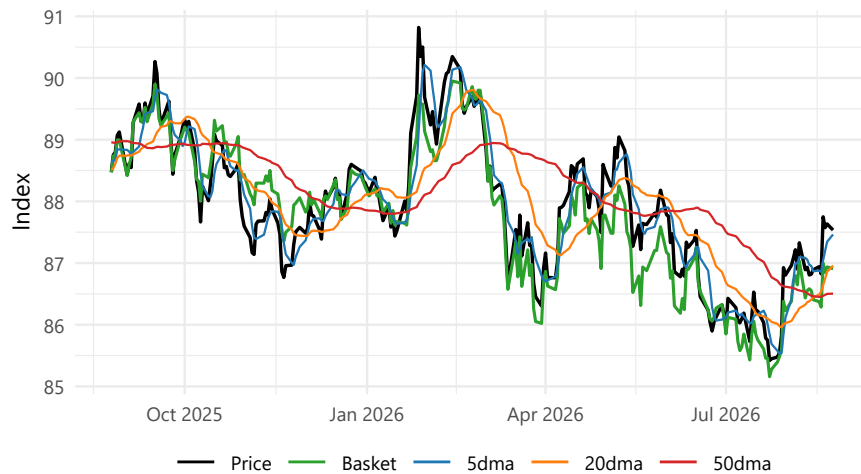
5	GSY	0.1100	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	LONG
8	BIL	0.1072	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	LONG
3	FTSM	0.1053	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	LONG
9	GBIL	0.0953	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	LONG
1	VNLA	0.0919	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	LONG

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-08-04 – filter **BLOCKED** (252d at date = -1.4%; needs > 0)
- Latest effective buy: 2026-05-29 – delayed re-arm from blocked cross (252d at fire = +0.6%)
- Latest sell_cross: 2026-07-23
- Current 252d return: -0.8% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-07-23 more recent than last effective buy 2026-05-29)



Category breakdown

category	n_members	abs_weight
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Currency	4	0.6409
International Fixed Income	2	0.3591

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.023

ticker	rep_wt	name	category	etf_signal
BWX	0.752	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	LONG
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	LONG
5	FXF	0.1651	Invesco CurrencyShares Swiss Franc Trust	Currency	LONG
3	FXB	0.1608	Invesco CurrencyShares British Pound St...	Currency	LONG
6	FXV	0.1448	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse

Members: 2 **Var explained:** 0.625 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-07-20 – filter **BLOCKED** (252d at date = -31.3%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-08-05
- Current 252d return: -34.2% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-08-05 more recent than last effective buy 2025-04-08)



Category breakdown

category	n_members	abs_weight
Leveraged / Inverse	1	0.5
Currency	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.022

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	FLAT

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
VHT	cluster_10	Healthcare	2026-05-19	66	19.5%	0.00	1.00
IYH	cluster_10	Healthcare	2026-05-20	65	19.3%	0.00	1.00
FHLC	cluster_10	Healthcare	2026-05-19	66	19.5%	0.00	1.00
XLV	cluster_10	Healthcare	2026-05-21	64	17.9%	0.00	1.00
PTH	cluster_10	Healthcare	2026-06-17	46	18.2%	0.00	1.00
PPH	cluster_10	Healthcare	2026-05-21	64	12.2%	0.00	1.00
GDX	cluster_17	Gold	2026-08-06	12	23.4%	0.00	1.00
IAU	cluster_17	Gold	2026-08-07	11	7.1%	0.00	1.00
GLD	cluster_17	Gold	2026-08-07	11	7.1%	0.00	1.00
SGOL	cluster_17	Gold	2026-08-07	11	7.1%	0.00	1.00
DBP	cluster_17	Gold	2026-08-07	11	7.3%	0.00	1.00
GDXJ	cluster_17	Gold	2026-08-06	12	22.1%	0.00	1.00
SIL	cluster_17	Gold	2026-08-07	11	13.4%	0.00	1.00
GOAU	cluster_17	Gold	2026-08-06	12	21.5%	0.00	1.00
GLTR	cluster_17	Gold	2026-08-10	10	5.3%	0.00	1.00
SLV	cluster_17	Gold	2026-08-10	10	4.7%	0.00	1.00
SIVR	cluster_17	Gold	2026-08-10	10	4.7%	0.00	1.00
SILJ	cluster_17	Gold	2026-08-07	11	9.8%	0.00	1.00
NANR	cluster_11	Materials	2026-07-24	21	12.2%	0.00	1.00
IGE	cluster_11	Materials	2026-07-23	22	9.7%	0.00	1.00
GNR	cluster_11	Materials	2026-07-27	20	10.9%	0.00	1.00
NORW	cluster_11	Materials	2026-08-05	13	6.5%	0.00	1.00
GUNR	cluster_11	Materials	2026-07-27	20	9.4%	0.00	1.00
MXI	cluster_11	Materials	2026-08-05	13	5.0%	0.00	1.00
COPX	cluster_11	Materials	2026-08-05	13	9.0%	0.00	1.00
PICK	cluster_11	Materials	2026-08-05	13	4.1%	0.01	0.99
XME	cluster_11	Materials	2026-08-10	10	0.3%	0.01	0.99
URA	cluster_11	Materials	2026-08-11	9	1.2%	0.01	0.99
EMLP	cluster_11	Materials	2026-08-13	7	-0.3%	0.23	0.60
MLPA	cluster_11	Materials	2026-07-10	31	4.2%	0.50	0.50
MLPX	cluster_11	Materials	2026-08-17	5	-0.8%	0.25	0.37
DBB	cluster_11	Materials	2026-08-04	14	0.9%	0.72	0.28
AMPL	cluster_11	Materials	2026-07-10	31	4.2%	0.86	0.14
SRLN	cluster_23	Bank Loan / Floating Ra...	2026-07-30	17	0.6%	0.00	1.00
BKLN	cluster_23	Bank Loan / Floating Ra...	2026-08-05	13	0.1%	0.00	1.00
FTSL	cluster_23	Bank Loan / Floating Ra...	2026-08-07	11	-0.3%	0.11	0.88
FLTR	cluster_23	Bank Loan / Floating Ra...	2026-08-21	1	-0.1%	0.17	0.49

FLOT	cluster_23	Bank Loan / Floating Ra...	2026-08-21	1	0.0%	0.23	0.39
FLRN	cluster_23	Bank Loan / Floating Ra...	2026-08-24	0	0.0%	0.32	0.10
FXB	cluster_25	International Fixed Inc...	2026-07-31	16	1.1%	0.00	1.00
FXE	cluster_25	International Fixed Inc...	2026-08-04	14	1.2%	0.00	1.00
FXF	cluster_25	International Fixed Inc...	2026-08-20	2	-0.3%	0.00	1.00
IGOV	cluster_25	International Fixed Inc...	2026-08-24	0	0.0%	0.03	0.97
SCHD	cluster_8	U.S. Dividend Equity	2026-07-07	34	8.2%	0.00	1.00
PWV	cluster_8	U.S. Dividend Equity	2026-04-02	98	18.3%	0.00	1.00
HDV	cluster_8	U.S. Dividend Equity	2026-06-25	41	8.7%	0.00	1.00
FDL	cluster_8	U.S. Dividend Equity	2026-07-16	27	4.3%	0.00	1.00
FTCS	cluster_8	U.S. Dividend Equity	2026-06-26	40	8.8%	0.02	0.98
FTA	cluster_8	U.S. Dividend Equity	2026-05-21	64	10.7%	0.02	0.98
ACWV	cluster_8	U.S. Dividend Equity	2026-07-06	35	4.3%	0.04	0.96
SPHD	cluster_8	U.S. Dividend Equity	2026-06-08	53	7.2%	0.05	0.95
SDY	cluster_8	U.S. Dividend Equity	2026-05-20	65	7.5%	0.06	0.94
OUSA	cluster_8	U.S. Dividend Equity	2026-07-01	37	6.5%	0.15	0.85
FVD	cluster_8	U.S. Dividend Equity	2026-06-10	51	6.9%	0.41	0.59
LGLV	cluster_8	U.S. Dividend Equity	2026-06-26	40	4.0%	0.59	0.40
USO	cluster_22	Broad Commodities	2026-08-11	9	3.6%	0.15	0.85
DBO	cluster_22	Broad Commodities	2026-08-11	9	3.4%	0.19	0.81
ARKK	cluster_15	Technology	2026-08-10	10	4.3%	0.00	1.00
ARKW	cluster_15	Technology	2026-08-21	1	-1.5%	0.00	1.00
SKYY	cluster_15	Technology	2026-07-30	17	13.5%	0.03	0.97
FDN	cluster_15	Technology	2026-08-03	15	1.3%	0.26	0.74
ARKQ	cluster_15	Technology	2026-08-11	9	-5.0%	0.17	0.71
IPO	cluster_15	Technology	2026-08-13	7	-6.3%	0.35	-0.15
GSY	cluster_24	Cash / Ultra-Short Fixe...	2026-08-07	11	-0.2%	0.00	1.00
NEAR	cluster_24	Cash / Ultra-Short Fixe...	2026-08-18	4	0.0%	0.00	1.00
FTSM	cluster_24	Cash / Ultra-Short Fixe...	2026-08-17	5	0.1%	0.05	0.91
ICSH	cluster_24	Cash / Ultra-Short Fixe...	2026-08-17	5	0.0%	0.14	0.78
MINT	cluster_24	Cash / Ultra-Short Fixe...	2026-08-17	5	0.1%	0.17	0.68
GBIL	cluster_24	Cash / Ultra-Short Fixe...	2026-08-18	4	0.1%	0.19	0.67
SHV	cluster_24	Cash / Ultra-Short Fixe...	2026-08-18	4	0.1%	0.20	0.65
BIL	cluster_24	Cash / Ultra-Short Fixe...	2026-08-18	4	0.1%	0.23	0.55
VNLA	cluster_24	Cash / Ultra-Short Fixe...	2026-08-21	1	-0.0%	0.12	0.54
EZA	cluster_7	Emerging Markets Equity	2026-08-05	13	6.7%	0.00	1.00
DVYE	cluster_7	Emerging Markets Equity	2026-07-24	21	4.8%	0.00	1.00
FEM	cluster_7	Emerging Markets Equity	2026-08-07	11	1.1%	0.01	0.99
EEMV	cluster_7	Emerging Markets Equity	2026-08-12	8	-0.1%	0.01	0.98
FEMS	cluster_7	Emerging Markets Equity	2026-08-12	8	0.7%	0.00	0.98
ECH	cluster_7	Emerging Markets Equity	2026-08-05	13	3.2%	0.02	0.98
DEM	cluster_7	Emerging Markets Equity	2026-08-07	11	0.7%	0.06	0.93
EMGF	cluster_7	Emerging Markets Equity	2026-08-14	6	-0.9%	0.02	0.87
EWX	cluster_7	Emerging Markets Equity	2026-08-14	6	-0.2%	0.01	0.87
EEMA	cluster_7	Emerging Markets Equity	2026-08-12	8	-1.1%	0.05	0.86

SPEM	cluster_7	Emerging Markets Equity	2026-08-06	12	0.2%	0.11	0.86
VWO	cluster_7	Emerging Markets Equity	2026-08-06	12	0.0%	0.12	0.86
ESGE	cluster_7	Emerging Markets Equity	2026-08-14	6	-0.8%	0.03	0.85
IEMG	cluster_7	Emerging Markets Equity	2026-08-13	7	-1.0%	0.03	0.84
SCHE	cluster_7	Emerging Markets Equity	2026-08-05	13	-0.0%	0.14	0.84
EEM	cluster_7	Emerging Markets Equity	2026-08-14	6	-0.8%	0.03	0.81
GEM	cluster_7	Emerging Markets Equity	2026-08-13	7	-1.0%	0.05	0.78
AAXJ	cluster_7	Emerging Markets Equity	2026-08-14	6	-1.3%	0.03	0.77
EWT	cluster_7	Emerging Markets Equity	2026-08-11	9	-0.6%	0.07	0.73
XSOE	cluster_7	Emerging Markets Equity	2026-08-17	5	-2.4%	0.04	0.69
FNDE	cluster_7	Emerging Markets Equity	2026-07-30	17	1.7%	0.34	0.66
AIA	cluster_7	Emerging Markets Equity	2026-08-14	6	-2.0%	0.11	0.61
SDIV	cluster_7	Emerging Markets Equity	2026-08-19	3	0.2%	0.39	0.60
GMF	cluster_7	Emerging Markets Equity	2026-08-06	12	-0.8%	0.25	0.58
PXH	cluster_7	Emerging Markets Equity	2026-07-31	16	0.7%	0.49	0.51
DGS	cluster_7	Emerging Markets Equity	2026-08-12	8	-1.4%	0.13	0.49
FXC	cluster_2	Developed ex-US Equity	2026-08-19	3	-0.3%	0.00	1.00
FXA	cluster_2	Developed ex-US Equity	2026-08-04	14	1.5%	0.00	1.00
EWC	cluster_2	Developed ex-US Equity	2026-07-10	31	5.9%	0.00	1.00
ENZL	cluster_2	Developed ex-US Equity	2026-07-07	34	6.8%	0.00	1.00
VIDI	cluster_2	Developed ex-US Equity	2026-08-10	10	1.6%	0.00	1.00
SCHC	cluster_2	Developed ex-US Equity	2026-08-05	13	2.3%	0.00	1.00
EWK	cluster_2	Developed ex-US Equity	2026-08-18	4	2.0%	0.01	0.99
EWS	cluster_2	Developed ex-US Equity	2026-06-17	46	14.0%	0.01	0.99
IQLT	cluster_2	Developed ex-US Equity	2026-07-01	37	6.1%	0.01	0.99
GWX	cluster_2	Developed ex-US Equity	2026-08-06	12	2.1%	0.01	0.99
FEP	cluster_2	Developed ex-US Equity	2026-07-31	16	2.6%	0.01	0.99
FNDF	cluster_2	Developed ex-US Equity	2026-07-31	16	3.0%	0.01	0.99
DLS	cluster_2	Developed ex-US Equity	2026-08-03	15	2.7%	0.01	0.99
VSS	cluster_2	Developed ex-US Equity	2026-08-07	11	0.2%	0.01	0.99
IQDF	cluster_2	Developed ex-US Equity	2026-08-04	14	1.2%	0.02	0.98
PXF	cluster_2	Developed ex-US Equity	2026-08-03	15	2.8%	0.02	0.98
IDV	cluster_2	Developed ex-US Equity	2026-07-28	19	3.5%	0.03	0.97
TLTD	cluster_2	Developed ex-US Equity	2026-07-30	17	3.0%	0.03	0.97
SPDW	cluster_2	Developed ex-US Equity	2026-08-05	13	1.1%	0.04	0.96
SCHF	cluster_2	Developed ex-US Equity	2026-08-05	13	1.0%	0.05	0.95
VEA	cluster_2	Developed ex-US Equity	2026-08-05	13	1.0%	0.05	0.95
QEFA	cluster_2	Developed ex-US Equity	2026-07-06	35	4.3%	0.05	0.95
JPIN	cluster_2	Developed ex-US Equity	2026-07-27	20	4.4%	0.06	0.94
VXUS	cluster_2	Developed ex-US Equity	2026-08-05	13	0.8%	0.06	0.94
IXUS	cluster_2	Developed ex-US Equity	2026-08-05	13	0.7%	0.06	0.94
ACWX	cluster_2	Developed ex-US Equity	2026-08-05	13	0.7%	0.06	0.94
VEU	cluster_2	Developed ex-US Equity	2026-08-05	13	0.8%	0.07	0.93
FNDC	cluster_2	Developed ex-US Equity	2026-08-04	14	1.2%	0.08	0.92
EWG	cluster_2	Developed ex-US Equity	2026-07-31	16	3.0%	0.08	0.92

NFRA	cluster_2	Developed	ex-US	Equity	2026-07-16	27	0.9%	0.09	0.91
CWI	cluster_2	Developed	ex-US	Equity	2026-08-04	14	0.5%	0.10	0.90
VGK	cluster_2	Developed	ex-US	Equity	2026-07-01	37	5.5%	0.11	0.89
EFAV	cluster_2	Developed	ex-US	Equity	2026-07-16	27	4.8%	0.12	0.88
IMTM	cluster_2	Developed	ex-US	Equity	2026-08-05	13	0.9%	0.10	0.88
EPOL	cluster_2	Developed	ex-US	Equity	2026-07-08	33	11.9%	0.12	0.88
INTF	cluster_2	Developed	ex-US	Equity	2026-07-27	20	4.0%	0.12	0.88
SCZ	cluster_2	Developed	ex-US	Equity	2026-08-04	14	1.3%	0.13	0.87
IEV	cluster_2	Developed	ex-US	Equity	2026-07-27	20	4.2%	0.13	0.87
IEUR	cluster_2	Developed	ex-US	Equity	2026-07-30	17	1.6%	0.14	0.86
EWL	cluster_2	Developed	ex-US	Equity	2026-07-28	19	1.6%	0.14	0.86
HDEF	cluster_2	Developed	ex-US	Equity	2026-07-09	32	5.4%	0.15	0.85
FEZ	cluster_2	Developed	ex-US	Equity	2026-07-30	17	2.7%	0.15	0.85
EZU	cluster_2	Developed	ex-US	Equity	2026-07-31	16	2.6%	0.16	0.84
EFG	cluster_2	Developed	ex-US	Equity	2026-08-05	13	0.3%	0.16	0.84
GSIE	cluster_2	Developed	ex-US	Equity	2026-07-02	36	4.7%	0.17	0.83
EWU	cluster_2	Developed	ex-US	Equity	2026-07-16	27	4.5%	0.17	0.83
IEFA	cluster_2	Developed	ex-US	Equity	2026-07-30	17	1.8%	0.17	0.83
RODM	cluster_2	Developed	ex-US	Equity	2026-07-27	20	2.2%	0.20	0.80
EFA	cluster_2	Developed	ex-US	Equity	2026-07-30	17	1.7%	0.20	0.80
IVLU	cluster_2	Developed	ex-US	Equity	2026-07-27	20	3.9%	0.22	0.78
HFXI	cluster_2	Developed	ex-US	Equity	2026-08-05	13	0.4%	0.18	0.78
EPP	cluster_2	Developed	ex-US	Equity	2026-07-15	28	4.4%	0.25	0.75
ESGD	cluster_2	Developed	ex-US	Equity	2026-07-30	17	1.6%	0.26	0.74
DWM	cluster_2	Developed	ex-US	Equity	2026-07-30	17	1.8%	0.27	0.73
EWN	cluster_2	Developed	ex-US	Equity	2026-08-05	13	0.6%	0.19	0.71
EWD	cluster_2	Developed	ex-US	Equity	2026-07-30	17	2.1%	0.30	0.70
VYMI	cluster_2	Developed	ex-US	Equity	2026-07-08	33	5.6%	0.32	0.68
EFV	cluster_2	Developed	ex-US	Equity	2026-07-08	33	6.0%	0.34	0.66
EW0	cluster_2	Developed	ex-US	Equity	2026-07-31	16	2.8%	0.32	0.64
BBAX	cluster_2	Developed	ex-US	Equity	2026-07-14	29	5.0%	0.39	0.61
EWA	cluster_2	Developed	ex-US	Equity	2026-07-20	25	5.4%	0.41	0.59
DWX	cluster_2	Developed	ex-US	Equity	2026-07-09	32	3.6%	0.49	0.51
EUFN	cluster_2	Developed	ex-US	Equity	2026-06-12	49	9.3%	0.50	0.50
EWQ	cluster_2	Developed	ex-US	Equity	2026-07-30	17	1.0%	0.62	0.38
HEZU	cluster_2	Developed	ex-US	Equity	2026-07-30	17	1.4%	0.54	0.35
VIGI	cluster_2	Developed	ex-US	Equity	2026-06-30	38	5.4%	0.67	0.33
IPKW	cluster_2	Developed	ex-US	Equity	2026-07-10	31	5.1%	0.69	0.31
IDLV	cluster_2	Developed	ex-US	Equity	2026-07-10	31	3.1%	0.87	0.13
VNQI	cluster_2	Developed	ex-US	Equity	2026-08-24	0	0.0%	0.95	-0.37
IYM	cluster_4	U.S.	Large	Cap Value	2026-08-06	12	5.8%	0.00	1.00
VTV	cluster_4	U.S.	Large	Cap Value	2026-04-13	92	11.9%	0.01	0.99
VLUE	cluster_4	U.S.	Large	Cap Value	2026-08-07	11	1.3%	0.01	0.98
FNDX	cluster_4	U.S.	Large	Cap Value	2026-04-09	94	13.1%	0.02	0.98
RSP	cluster_4	U.S.	Large	Cap Value	2026-04-14	91	11.2%	0.03	0.97

VAW	cluster_4	U.S. Large Cap Value	2026-08-06	12	2.7%	0.04	0.96
PKW	cluster_4	U.S. Large Cap Value	2026-05-20	65	11.6%	0.05	0.95
PRF	cluster_4	U.S. Large Cap Value	2026-04-09	94	13.5%	0.06	0.94
DGRO	cluster_4	U.S. Large Cap Value	2026-04-13	92	10.6%	0.08	0.92
DTD	cluster_4	U.S. Large Cap Value	2026-04-13	92	9.7%	0.08	0.92
DLN	cluster_4	U.S. Large Cap Value	2026-04-13	92	9.9%	0.09	0.91
VONV	cluster_4	U.S. Large Cap Value	2026-04-10	93	16.4%	0.10	0.90
IWD	cluster_4	U.S. Large Cap Value	2026-04-10	93	16.4%	0.10	0.90
RSPM	cluster_4	U.S. Large Cap Value	2026-07-27	20	4.5%	0.12	0.88
XLB	cluster_4	U.S. Large Cap Value	2026-07-28	19	2.4%	0.12	0.88
SCHV	cluster_4	U.S. Large Cap Value	2026-08-04	14	-0.1%	0.12	0.87
IWX	cluster_4	U.S. Large Cap Value	2026-04-10	93	17.6%	0.13	0.87
JVAL	cluster_4	U.S. Large Cap Value	2026-07-30	17	3.0%	0.23	0.77
FEX	cluster_4	U.S. Large Cap Value	2026-04-09	94	10.0%	0.22	0.77
RWL	cluster_4	U.S. Large Cap Value	2026-04-10	93	13.6%	0.30	0.70
VIG	cluster_4	U.S. Large Cap Value	2026-04-14	91	9.2%	0.34	0.66
IYZ	cluster_4	U.S. Large Cap Value	2026-08-06	12	-1.8%	0.37	0.62
ITB	cluster_4	U.S. Large Cap Value	2026-07-28	19	-2.5%	0.58	-0.44
AIVL	cluster_4	U.S. Large Cap Value	2026-04-10	93	11.4%		
TUR	cluster_18	India Equity	2026-08-14	6	2.2%	0.00	1.00
EWM	cluster_18	India Equity	2026-07-30	17	1.2%	0.01	0.99
FV	cluster_1	U.S. Large Cap Equity	2026-08-10	10	0.8%	0.00	1.00
QQEW	cluster_1	U.S. Large Cap Equity	2026-08-04	14	0.5%	0.02	0.98
XT	cluster_1	U.S. Large Cap Equity	2026-08-07	11	-0.9%	0.08	0.89
AOM	cluster_1	U.S. Large Cap Equity	2026-08-06	12	0.1%	0.11	0.88
XLC	cluster_1	U.S. Large Cap Equity	2026-08-06	12	1.0%	0.12	0.88
SPGM	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.1%	0.16	0.84
DSI	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.8%	0.16	0.84
VT	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.2%	0.16	0.84
AOA	cluster_1	U.S. Large Cap Equity	2026-08-05	13	-0.3%	0.17	0.83
AOK	cluster_1	U.S. Large Cap Equity	2026-08-06	12	0.0%	0.12	0.82
AOR	cluster_1	U.S. Large Cap Equity	2026-08-05	13	-0.2%	0.17	0.82
SUSA	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.6%	0.18	0.82
IYC	cluster_1	U.S. Large Cap Equity	2026-08-04	14	0.5%	0.18	0.81
EIS	cluster_1	U.S. Large Cap Equity	2026-08-12	8	-1.4%	0.17	0.81
ACWI	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.3%	0.19	0.81
QAI	cluster_1	U.S. Large Cap Equity	2026-08-06	12	0.2%	0.14	0.80
CRBN	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.4%	0.19	0.80
XLY	cluster_1	U.S. Large Cap Equity	2026-08-05	13	-0.3%	0.19	0.78
DGRW	cluster_1	U.S. Large Cap Equity	2026-08-03	15	1.5%	0.23	0.77
URTH	cluster_1	U.S. Large Cap Equity	2026-08-03	15	1.3%	0.24	0.76
VONE	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.9%	0.25	0.74
IWV	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.9%	0.26	0.74
IWB	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.9%	0.26	0.73
EPS	cluster_1	U.S. Large Cap Equity	2026-08-03	15	0.8%	0.27	0.73

VTHR	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.9%	0.27	0.73
ILCB	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.9%	0.27	0.72
IWP	cluster_1	U.S. Large Cap Equity	2026-08-07	11	-1.6%	0.19	0.72
IYY	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.28	0.71
VTI	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.28	0.71
VV	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.28	0.71
ITOT	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.29	0.71
SCHB	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-0.9%	0.29	0.70
IVV	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.30	0.70
SPY	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.30	0.69
VOO	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.30	0.69
SCHX	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.30	0.69
MGC	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.1%	0.30	0.69
IWL	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.1%	0.30	0.69
GSLC	cluster_1	U.S. Large Cap Equity	2026-08-03	15	0.9%	0.32	0.68
FDIS	cluster_1	U.S. Large Cap Equity	2026-08-05	13	-0.8%	0.26	0.68
LRGF	cluster_1	U.S. Large Cap Equity	2026-08-03	15	1.0%	0.32	0.67
VCR	cluster_1	U.S. Large Cap Equity	2026-08-05	13	-0.9%	0.26	0.67
OEF	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.3%	0.32	0.67
JHML	cluster_1	U.S. Large Cap Equity	2026-08-03	15	0.8%	0.32	0.67
SPTM	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.1%	0.32	0.67
ESGU	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.1%	0.32	0.67
CATH	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.2%	0.34	0.65
SPYX	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.2%	0.38	0.60
PRNT	cluster_1	U.S. Large Cap Equity	2026-08-05	13	-2.4%	0.40	0.58
VOT	cluster_1	U.S. Large Cap Equity	2026-08-05	13	-1.1%	0.32	0.57
IOO	cluster_1	U.S. Large Cap Equity	2026-08-03	15	0.0%	0.45	0.55
XLG	cluster_1	U.S. Large Cap Equity	2026-08-05	13	-1.8%	0.41	0.52
IMCG	cluster_1	U.S. Large Cap Equity	2026-08-04	14	-1.0%	0.43	0.38
SPHB	cluster_1	U.S. Large Cap Equity	2026-08-12	8	-2.9%	0.19	0.38
FCOM	cluster_1	U.S. Large Cap Equity	2026-08-13	7	-0.6%	0.35	0.33
VOX	cluster_1	U.S. Large Cap Equity	2026-08-13	7	-0.6%	0.35	0.30
VFMO	cluster_1	U.S. Large Cap Equity	2026-08-13	7	-3.2%	0.22	0.13
RPG	cluster_1	U.S. Large Cap Equity	2026-08-17	5	-6.0%	0.26	-0.03
SPHQ	cluster_1	U.S. Large Cap Equity	2026-08-14	6	-1.9%	0.24	-0.03
CWB	cluster_1	U.S. Large Cap Equity	2026-08-14	6	-3.7%	0.40	-0.08
ICVT	cluster_1	U.S. Large Cap Equity	2026-08-17	5	-3.7%	0.34	-0.11
FTC	cluster_1	U.S. Large Cap Equity	2026-08-10	10	-2.7%	0.49	-0.25
PDP	cluster_1	U.S. Large Cap Equity	2026-08-17	5	-7.5%	0.44	-0.35
MTUM	cluster_1	U.S. Large Cap Equity	2026-08-18	4	-3.6%	0.47	-0.41
QMOM	cluster_1	U.S. Large Cap Equity	2026-08-12	8	-5.1%	0.69	-0.48
XMMO	cluster_1	U.S. Large Cap Equity	2026-08-17	5	-6.9%	0.70	-0.64
VPL	cluster_21	Japan Equity	2026-08-10	10	1.2%	0.08	0.83
IPAC	cluster_21	Japan Equity	2026-08-03	15	2.2%	0.40	0.59
BBJP	cluster_21	Japan Equity	2026-08-05	13	-0.3%	0.44	0.44

EWJ	cluster_21	Japan Equity	2026-08-05	13	-0.3%	0.45	0.42
EWJV	cluster_21	Japan Equity	2026-07-06	35	2.2%	0.89	0.10
DBJP	cluster_21	Japan Equity	2026-08-10	10	-1.2%	0.51	0.00
FLJH	cluster_21	Japan Equity	2026-08-10	10	-0.9%	0.49	-0.04
HEWJ	cluster_21	Japan Equity	2026-08-07	11	-1.1%	0.54	-0.09
DXJ	cluster_21	Japan Equity	2026-08-05	13	0.5%	0.73	-0.26
KCE	cluster_13	Financials	2026-07-07	34	8.3%	0.00	1.00
XLF	cluster_13	Financials	2026-04-13	92	12.7%	0.72	0.28
FNCL	cluster_13	Financials	2026-04-13	92	12.5%	0.73	0.27
KURE	cluster_12	China Equity	2026-07-06	35	8.6%	0.00	1.00
GXC	cluster_12	China Equity	2026-07-31	16	-1.1%	0.78	0.22
CXSE	cluster_12	China Equity	2026-08-07	11	-4.2%	0.67	-0.05
REM	cluster_3	U.S. Mid Cap Equity	2026-08-12	8	0.3%	0.01	0.99
SMMV	cluster_3	U.S. Mid Cap Equity	2026-06-25	41	5.9%	0.03	0.97
IWC	cluster_3	U.S. Mid Cap Equity	2026-08-05	13	-0.0%	0.05	0.93
VBK	cluster_3	U.S. Mid Cap Equity	2026-08-07	11	-1.6%	0.11	0.82
SCHM	cluster_3	U.S. Mid Cap Equity	2026-08-07	11	-0.8%	0.11	0.77
SMLF	cluster_3	U.S. Mid Cap Equity	2026-08-03	15	1.0%	0.23	0.75
ISCG	cluster_3	U.S. Mid Cap Equity	2026-08-04	14	-1.1%	0.23	0.73
VXF	cluster_3	U.S. Mid Cap Equity	2026-08-05	13	-0.2%	0.23	0.72
VTWO	cluster_3	U.S. Mid Cap Equity	2026-08-04	14	-1.2%	0.26	0.64
IWM	cluster_3	U.S. Mid Cap Equity	2026-08-04	14	-1.2%	0.27	0.62
IWO	cluster_3	U.S. Mid Cap Equity	2026-08-06	12	-0.3%	0.19	0.61
VTWG	cluster_3	U.S. Mid Cap Equity	2026-08-06	12	-0.3%	0.20	0.60
SCHA	cluster_3	U.S. Mid Cap Equity	2026-08-07	11	-1.0%	0.19	0.42
IVOG	cluster_3	U.S. Mid Cap Equity	2026-08-06	12	-0.8%	0.44	0.33
IJK	cluster_3	U.S. Mid Cap Equity	2026-08-06	12	-0.7%	0.44	0.31
MDYG	cluster_3	U.S. Mid Cap Equity	2026-08-06	12	-0.7%	0.45	0.31
IJS	cluster_3	U.S. Mid Cap Equity	2026-04-10	93	12.2%	0.70	0.24
IYT	cluster_3	U.S. Mid Cap Equity	2026-08-06	12	0.7%	0.57	-0.49
QYLD	cluster_6	U.S. Large Cap Growth	2026-08-07	11	-0.4%	0.01	0.97
AIQ	cluster_6	U.S. Large Cap Growth	2026-08-10	10	-1.9%	0.10	0.81
FXL	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-0.4%	0.20	0.77
SCHG	cluster_6	U.S. Large Cap Growth	2026-08-04	14	-1.0%	0.26	0.74
PTLC	cluster_6	U.S. Large Cap Growth	2026-08-04	14	-1.0%	0.30	0.69
ONEQ	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-1.4%	0.31	0.57
MGK	cluster_6	U.S. Large Cap Growth	2026-08-05	13	-1.9%	0.35	0.57
TDIV	cluster_6	U.S. Large Cap Growth	2026-08-07	11	-3.1%	0.38	0.56
VUG	cluster_6	U.S. Large Cap Growth	2026-08-05	13	-2.0%	0.40	0.53
NXTG	cluster_6	U.S. Large Cap Growth	2026-08-12	8	-3.2%	0.28	0.51
IUSG	cluster_6	U.S. Large Cap Growth	2026-08-05	13	-2.3%	0.43	0.46
VOOG	cluster_6	U.S. Large Cap Growth	2026-08-05	13	-2.4%	0.43	0.46
SPYG	cluster_6	U.S. Large Cap Growth	2026-08-05	13	-2.4%	0.44	0.46
IVW	cluster_6	U.S. Large Cap Growth	2026-08-05	13	-2.4%	0.44	0.46
VONG	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-2.0%	0.36	0.44

VGT	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-2.7%	0.41	0.44
IWF	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-2.0%	0.36	0.43
FTEC	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-2.7%	0.41	0.42
QQQ	cluster_6	U.S. Large Cap Growth	2026-08-07	11	-2.3%	0.26	0.42
PTNQ	cluster_6	U.S. Large Cap Growth	2026-08-07	11	-2.3%	0.26	0.42
IWY	cluster_6	U.S. Large Cap Growth	2026-08-07	11	-2.8%	0.36	0.40
IXN	cluster_6	U.S. Large Cap Growth	2026-08-07	11	-2.9%	0.30	0.39
QTEC	cluster_6	U.S. Large Cap Growth	2026-08-10	10	-1.9%	0.23	0.36
IYW	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-2.6%	0.42	0.30
XNTK	cluster_6	U.S. Large Cap Growth	2026-08-13	7	-4.8%	0.20	0.28
XLK	cluster_6	U.S. Large Cap Growth	2026-08-07	11	-4.2%	0.40	0.25
QTUM	cluster_6	U.S. Large Cap Growth	2026-08-13	7	-7.1%	0.28	0.20
ILCG	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-2.6%	0.49	0.16
IGM	cluster_6	U.S. Large Cap Growth	2026-08-06	12	-3.1%	0.51	0.10
NULG	cluster_6	U.S. Large Cap Growth	2026-08-05	13	-2.9%	0.55	-0.01
PTF	cluster_6	U.S. Large Cap Growth	2026-04-07	96	5.5%	0.48	-0.42
PWB	cluster_6	U.S. Large Cap Growth	2026-08-07	11	-3.4%	0.60	-0.45
BSCQ	cluster_5	Aggregate Fixed Income	2026-08-07	11	-0.2%	0.00	1.00
VGSH	cluster_5	Aggregate Fixed Income	2026-08-17	5	-0.0%	0.01	0.98
SHY	cluster_5	Aggregate Fixed Income	2026-08-17	5	-0.0%	0.02	0.98
SCHO	cluster_5	Aggregate Fixed Income	2026-08-17	5	-0.0%	0.02	0.96
SPSB	cluster_5	Aggregate Fixed Income	2026-08-18	4	0.0%	0.02	0.95
LDUR	cluster_5	Aggregate Fixed Income	2026-08-17	5	-0.0%	0.05	0.94
SPTS	cluster_5	Aggregate Fixed Income	2026-08-17	5	0.0%	0.04	0.93
LMBS	cluster_5	Aggregate Fixed Income	2026-08-19	3	-0.4%	0.02	0.77
IBDR	cluster_5	Aggregate Fixed Income	2026-08-20	2	0.0%	0.23	0.54
EMLC	cluster_9	High Yield Bond	2026-08-05	13	0.9%	0.00	1.00
EBND	cluster_9	High Yield Bond	2026-07-31	16	1.5%	0.00	1.00
SHYG	cluster_9	High Yield Bond	2026-08-19	3	0.0%	0.00	0.90
HYG	cluster_9	High Yield Bond	2026-08-19	3	-0.0%	0.01	0.71
PFXF	cluster_9	High Yield Bond	2026-08-10	10	-0.8%	0.31	0.65
HYLB	cluster_9	High Yield Bond	2026-08-19	3	0.0%	0.03	0.56
HYLS	cluster_9	High Yield Bond	2026-08-13	7	-0.7%	0.16	0.44
PFF	cluster_9	High Yield Bond	2026-08-13	7	-1.2%	0.67	-0.07
PBJ	cluster_19	Consumer Staples	2026-08-18	4	1.9%	0.03	0.95
PSCC	cluster_19	Consumer Staples	2026-06-10	51	11.2%	0.11	0.89
XLP	cluster_19	Consumer Staples	2026-07-27	20	2.4%	0.13	0.86
FSTA	cluster_19	Consumer Staples	2026-07-28	19	0.1%	0.21	0.78
VDC	cluster_19	Consumer Staples	2026-07-28	19	0.1%	0.22	0.77
SUB	cluster_16	Municipal Bond	2026-08-13	7	0.0%	0.01	0.97
XLRE	cluster_14	Real Estate	2026-08-18	4	1.6%	0.73	-0.26
VNQ	cluster_14	Real Estate	2026-08-19	3	0.5%	0.77	-0.48
FREL	cluster_14	Real Estate	2026-08-19	3	0.4%	0.79	-0.51
REZ	cluster_14	Real Estate	2026-08-24	0	0.0%	0.76	-0.63
REET	cluster_14	Real Estate	2026-08-18	4	1.1%	0.97	-0.93

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
IBB	cluster_10	Healthcare	1.00	1.00
FXC	cluster_2	Developed ex-US Equity	1.00	1.00
NANR	cluster_11	Materials	1.00	1.00
GDX	cluster_17	Gold	1.00	1.00
SCHD	cluster_8	U.S. Dividend Equity	1.00	1.00
EMLC	cluster_9	High Yield Bond	1.00	1.00
ARKK	cluster_15	Technology	1.00	1.00
GSY	cluster_24	Cash / Ultra-Short Fixed ...	1.00	1.00
EZA	cluster_7	Emerging Markets Equity	1.00	1.00
SRLN	cluster_23	Bank Loan / Floating Rate	1.00	1.00
IYM	cluster_4	U.S. Large Cap Value	1.00	1.00
COMT	cluster_22	Broad Commodities	1.00	1.00
FV	cluster_1	U.S. Large Cap Equity	1.00	1.00
BSCQ	cluster_5	Aggregate Fixed Income	1.00	1.00
KCE	cluster_13	Financials	1.00	1.00
REM	cluster_3	U.S. Mid Cap Equity	1.00	0.99
SUB	cluster_16	Municipal Bond	0.98	0.97
QYLD	cluster_6	U.S. Large Cap Growth	0.98	0.97
PBJ	cluster_19	Consumer Staples	0.98	0.95
VPL	cluster_21	Japan Equity	0.91	0.83

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
PSK	cluster_9	High Yield Bond	1.00	-1.00
UUP	cluster_26	Leveraged / Inverse	1.00	-1.00
XTN	cluster_3	U.S. Mid Cap Equity	1.00	-1.00
IGF	cluster_2	Developed ex-US Equity	1.00	-1.00
BNDX	cluster_5	Aggregate Fixed Income	0.97	-0.97
IAT	cluster_13	Financials	1.00	-0.96

EWZ	cluster_7	Emerging Markets Equity	0.98	-0.95
PAVE	cluster_4	U.S. Large Cap Value	0.98	-0.94
RWO	cluster_14	Real Estate	0.97	-0.94
XLU	cluster_19	Consumer Staples	0.93	-0.93
CQQQ	cluster_12	China Equity	0.92	-0.92
PRN	cluster_1	U.S. Large Cap Equity	0.92	-0.92
IGHG	cluster_23	Bank Loan / Floating Rate	0.91	-0.91
EPHE	cluster_18	India Equity	1.00	-0.91
PSCT	cluster_6	U.S. Large Cap Growth	0.91	-0.90
SPLV	cluster_8	U.S. Dividend Equity	0.97	-0.88
HYMB	cluster_16	Municipal Bond	0.74	-0.74
DXJ	cluster_21	Japan Equity	0.73	-0.26
TDDT	cluster_20	Inflation-Protected Fixed...	0.22	-0.19
IPO	cluster_15	Technology	0.35	-0.15

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_18	12	FLAT	0.487	0.072	0.00016	
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_18	12	FLAT	0.302	0.094	0.00021	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.730	0.096	0.00024	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.647	0.206	0.00036	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.759	0.028	0.00007	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_18	12	FLAT	0.591	0.168	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.734	0.120	0.00031	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.734	0.141	0.00039	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.715	0.181	0.00044	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.694	0.166	0.00045	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.699	0.187	0.00046	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_18	12	LONG	0.511	0.216	0.00055	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_5	58	LONG	0.847	0.125	0.00037	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.883	0.026	0.00001	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	LONG	0.883	0.054	0.00018	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_5	58	FLAT	0.877	0.167	0.00032	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	LONG	0.820	0.202	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_5	58	LONG	0.833	0.239	0.00049	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_18	12	FLAT	0.447	0.270	0.00074	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	FLAT	0.878	0.250	0.00047	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_18	12	FLAT	0.445	0.328	0.00083	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_12	15	LONG	0.781	0.276	0.00072	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.878	0.265	0.00055	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_11	17	LONG	0.742	0.352	0.00084	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_18	12	FLAT	0.501	0.350	0.00102	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_12	15	FLAT	0.853	0.255	0.00073	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	LONG	0.867	0.322	0.00071	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_18	12	FLAT	0.559	0.408	0.00108	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_12	15	FLAT	0.903	0.237	0.00064	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_20	10	FLAT	0.919	0.233	0.00059	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20